MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, TTL, MONOSTABLE MULTIVIBRATORS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for monolithic silicon, TTL, monostable multivibrator microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
01	Single monostable multivibrator, with Schmitt trigger input
02	Single retriggerable monostable multivibrator with clear
03	Dual retriggerable monostable multivibrator with clear
04	One shot multivibrator
05	Dual one shot multivibrator

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Letter	Case outline, (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat-package
B	F-3 (14-lead, 3/16" x 1/4"), flat-package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat-package
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat-package

1.3 Absolute maximum ratings.

Supply voltage range - - - - - - -	-0.5 V dc to 7.0 V dc
Input voltage range- - - - - - - -	-1.5 V dc at -12 mA to 5.5 V dc
Storage temperature range- - - - -	-65°C to +150°C
Maximum power dissipation (P_D), per multivibrator 1/ - - - - - - - -	220 mW for 01; 170 mW for 02; 190 mW for 03; 138 mW for 04; and 143 mW for 05

1/ Must withstand the added P_D due to short circuit test (e.g. I_{OS}).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, (RSE-2) Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases A, B, C, D, E, F	(See MIL-M-38510, appendix C)
Junction temperature (T_J) 2/	+175°C

1.4 Recommended operating conditions.

Supply voltage(V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage (V_{IH})	2.4 V dc
Maximum low level input voltage (V_{IL})	.4 V dc
Normalized fanout (each output) 3/	10 maximum (device type 01) 10 maximum (device types 02 and 03, low level logic) 20 maximum (device types 02 and 03, high level logic) 6 maximum (device type 04, low level logic) 12 maximum (device type 04, high level logic) 8 maximum (device type 05, low level logic) 16 maximum (device type 05, high level logic)
Case operating temperature range (T_C)	-55°C to +125°C
Input pulse rise/fall time, (device type 01)	
Schmitt input (B)	1 V/s maximum
Positive gains threshold voltage (V_{T+})	2.0 V maximum
Negative gains threshold voltage (V_{T-})	0.8 V maximum
Logic inputs (A1,A2)	1 V/µs maximum
Input data setup time, (t_{SETUP})	
Device type 01	60 ns minimum
Device types 02, 03, 04, and 05	40 ns minimum
Input data hold time, (t_{HOLD})	
Device type 01	0
Device types 02, 03, 04, and 05	40 ns minimum
External timing resistance	
Device type 01	30 kΩ maximum
Device types 02, 03, 04, and 05	5 kΩ minimum to 25 kΩ maximum

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

2/ Maximum junction temperature shall not be exceeded except for short duration burn-in screening conditions per method 5004 of MIL-STD-883.
3/ The device will fanout in both high and low levels to the specified number of data inputs on the same device as that being tested.

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STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS srandards), the text of this specification shall take precedence. Nothing in this specificaton, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables. The truth tables shall be as specified on figure 2.

3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.4 Case outlines. The case outlines shall be in accordance with 1.2.3 herein.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements shall be as specified in table III for the applicable device type and device class. The subgroups of table III which constitute the minimum electrical test requirements for screening, qualification, and quality conformance by device class, are specified in table II.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 3 (see MIL-M-38510, appendix E).

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C < T_C < +125°C$ unless otherwise specified	Device type	Limits Min	Limits Max	Unit
High level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -.4 mA	01	2.4		V
		V_{CC} = 4.5 V, I_{OH} = -.8 mA	02,03 1/	2.4		V
		V_{CC} = 4.5 V, I_{OH} = -0.72 mA	04	2.4		V
		V_{CC} = 4.5 V, I_{OH} = -0.96 mA	05	2.4		V
Low level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 16 mA	01, 02, 03 1/		0.4	V
		V_{CC} = 4.5 V, I_{OL} = 10 mA	04		0.4	V
		V_{CC} = 4.5 V, I_{OL} = 10 mA	05		0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -12 mA, T_C = +25°C	All		-1.5	V
Low level input current at A_1 or A_2	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	01	-.7	-1.6	mA
Low level input current at B	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	01	-1.4	-3.2	mA
Low level input current at data inputs	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	02, 03	-.7	-1.6	mA
Low level input current at clear input	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	02, 03	-1.4	-3.2	mA
Low level input current at data inputs	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	04	-.7	-1.6	mA
Low level input current at data and clear inputs	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	05	-.7	-1.6	mA
High level input current at A_1 or A_2	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	01		40	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	01		100	µA
High level input current at 3	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	01		80	µA
	I_{IH4}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	01		200	µA
High level input current at data input	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	02, 03		40	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	02, 03		100	µA
High level input current at clear input	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	02, 03		80	µA
	I_{IH4}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	02, 03		200	µA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified	Device type	Limits Min	Limits Max	Unit
High level input current at data inputs	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	04		40	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	04		100	µA
High level input current at data and clear inputs	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	05		40	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	05		100	µA
Short circuit output current	I_{OS}	V_{CC} = 5.5 V, V_{IN} = 0 V 1/ 2/	01	-20	-55	mA
			02, 03	-10	-40	mA
			04	-10	-40	mA
			05	-10	-40	mA
Supply current (quiescent)	I_{CC1}	V_{CC} = 5.5 V 3/	01		25	mA
			02		28	mA
			03		66	mA
			04		25	mA
			05		52	mA
Supply current (triggered)	I_{CC2}	V_{CC} = 5.5 V 4/	01		40	mA
			02		28	mA
			03		66	mA
Propagation delay time to high level (B input to Q output)	t_{PLH1}	V_{CC} = 5.0 V; C_L = 50 pF ±10%; C_X = 80 pF ±10%; R_L = 390Ω ±5% Figure 4 (device type 01)	01	15	75	ns
Propagation delay time to low level (B input to Q output)	t_{PHL1}			20	87	ns
Propagation delay time to high level (A_1 or A_2 inputs to Q output)	t_{PLH2}			25	93	ns
Propagation delay time to low level (A_1 or A_2 inputs to $\overline{Q}$ output)	t_{PHL2}			30	106	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified	Device type	Limits Min	Limits Max	Unit
Propagation delay time to high level (A_1 or A_2 inputs to Q output)	t_{PLH1}		02, 03	7	48	ns
Propagation delay time to high level (B_1 or B_2 inputs to Q output)	t_{PLH2}	V_{CC} = 5.0 V; C_X = 1,000 pF ±10%; C_L = 50 pF ±10%; R_X = 10 kΩ ±5%;		7	41	ns
Propagation delay time to low level (A_1 or A_2 inputs to $\overline{Q}$ output)	t_{PHL1}	R_L = 390Ω ±5% Figure 6 (device type 02) Figure 7 (device type 03)		7	56	ns
Propagation delay time to low level (B_1 or B_2 inputs to $\overline{Q}$ output)	t_{PHL2}			7	51	ns
Propagation delay time to low level (clear input to Q output)	t_{PHL3}			7	39	ns
Propagation delay time to high level (clear input to $\overline{Q}$ output)	t_{PLH3}			7	56	ns
Minimum pulse width of Q output pulse	$t_{W(MIN)}$	C_X = 0; R_X = 5 kΩ ±5%			75	ns
		C_X = 15 pF (max); R_X = 5 kΩ ±5% 5/			105	ns
Width of Q output pulse	t_W			2.50	4.15	µs
Propagation delay time to high level (A_1 or A_2 inputs to Q output)	t_{PLH1}		04	7	48	ns
Propagation delay time to high level (B_1 or B_2 inputs to Q output)	t_{PLH2}			7	41	ns
Propagation delay time to low level (A_1 or A_2 inputs to $\overline{Q}$ output)	t_{PHL1}	V_{CC} = 5.0 V; C_X = Open; C_L = 50 pF ±10%; R_X = 5 kΩ ±5%; R_L = 390Ω ±5%		7	55	ns
Propagation delay time to low level (B_1 or B_2 inputs to $\overline{Q}$ output)	t_{PHL2}	Figure 8 (device type 04)		7	51	ns
Minimum pulse width of Q output pulse	$t_{W(MIN)}$	C_X = 0; R_X = 5 kΩ ±5%		25	95	ns
		C_X = 15 pF (max); R_X = 5 kΩ ±5% 5/		25	125	ns
Width of Q output pulse	t_d	R_L = 390Ω ±5%; V_{CC} = 5.0 V; C_L = 50 pF ±10% C_X = 1,000 pF ±10%; R_X = 10 kΩ ±5% Figure 8 (device type 04)	04	2.60	4.10	µs

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C < T_C < +125°C$ unless otherwise specified	Device type	Limits Min	Limits Max	Unit
Propagation delay time to high level (A input to Q output)	t_{PLH1}	V_{CC} = 5.0 V; C_X = Open; C_L = 50 pF ±10%; R_X = 10 kΩ ±5%; R_L = 390Ω ±5% Figure 9 (device type 05)	05	7	54	ns
Propagation delay time to high level (B input to Q output)	t_{PLH2}			7	51	ns
Propagation delay time to low level (A input to Q output)	t_{PHL1}			7	61	ns
Propagation delay time to low level (B input to Q output)	t_{PHL2}			7	58	ns
Propagation delay time to low level (clear input to Q output)	t_{PHL3}			7	39	ns
Propagation delay time to high level (clear input to Q output)	t_{PLH3}			7	56	ns
Minimum pulse width of Q output pulse	$t_{W(MIN)}$	C_X = 0;		35	108	ns
		C_X = 15 pF (max) 5/		35	140	ns
Width of Q output pulse	t_W	R_L = 390Ω ±5%; V_{CC} = 5.0 V; C_L = 50 pF ±10%; C_X = 1,000 pF ±10%; R_X = 10 kΩ ±5% Figure 9 (device type 05)	05	2.60	3.91	s
Pulse width obtained with internal timing resistor	$t_{P(OUT)1}$	R_L = 390Ω ±5%; V_{CC} = 5.0 V; C_L = 50 pF ±10%; R_X = Open (figure 6); C_X = 80 pF ±10%; pin 9 connected to V_{CC}	01	70	150	ns
	$t_{P(OUT)2}$	R_L = 390Ω ±5%; V_{CC} = 5.0 V; C_L = 50 pF ±10%; R_X = Open; C_X = 15 pF ±10%; pin 9 connected to V_{CC} Figure 5	01	20	50	ns
Pulse width obtained with external timing resistor	$t_{P(OUT)3}$	V_{CC} = 5 V; C_X = 100 pF ±10% C_L = 50 pF ±10%; R_X = 10 kΩ ±5%	01	600	825	ns
	$t_{P(OUT)4}$	pin 9 open; C_X = 1,000 pF ±10% Figure 5	01	5.5	8	ms

1/ Ground C_X to measure V_{OH} at Q, V_{OL} at $\overline{Q}$, or I_{OS} at Q. C_X is open to measure V_{OH} at $\overline{Q}$, V_{OL} at Q, or I_{OS} at $\overline{Q}$. (Types 02 and 03.)
2/ Not more than one output should be shorted at a time.
3/ For types 02 and 03: I_{CC} is measured (after clearing) with 2.4 V applied to all clear and A inputs, B inputs grounded, all outputs open, C_X = 0.02 µF and R_X = 25 kΩ. R_I of type 02 is open.
4/ For types 02 and 03: I_{CC} is measured in the triggered state with 2.4 V applied to all clear and B inputs, A inputs grounded, all outputs open, C_X = 0.02 µF and R_X = 25 kΩ. R_I of type 02 is open.
5/ 15 pF load is for automatic test equipment only, which includes probe and jig capacitance.

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Initial electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7,9, 10,11	1*,2,3,7,9
Group A test requirements (method 5005)	1,2,3,7,8, 9,10,11	1,2,3,7,8,9
Group B test requirements (method 5005) subgroup 5	1,2,3,9,10, 11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Additional electrical subgroups for group C periodic inspections	N/A	10,11
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

1. Test condition D or E, using the circuit shown on figure 10, or equivalent.

2. T_A = +125°C minimum.

b. Initial and final electrical test parameters shall be as specified in table II, except initial electrical tests prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for group A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

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4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions or equivalent.

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C, minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

c. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment, design application and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

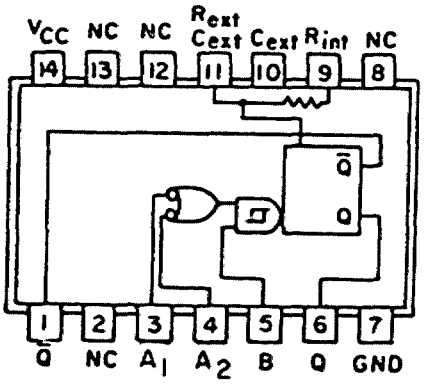
g. Requirements for special carriers, lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

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Device type 01

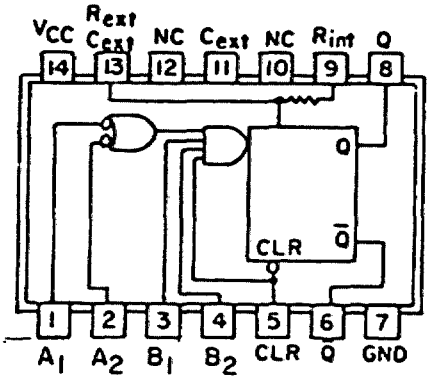
Cases A, B, C, and D



NOTES (device type 01):
1. To use internal timing resistor (2 kΩ nom.) connect pin 9 (R_I) to pin 14 (V_{CC}).
2. To obtain a variable pulse width, connect external variable resistor between pins 9 (R_I) and 14 (V_{CC}). No external current limiting is required.
3. External timing capacitor may be connected between pins 10 (positive) (C_X) and 11 (R_X/C_X).

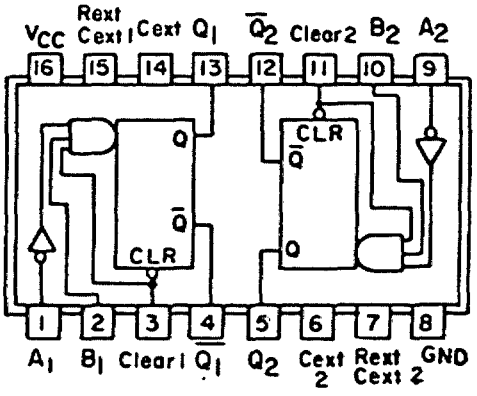
Device type 02

Cases A, B, C, and D



Device type 03

Case E and F



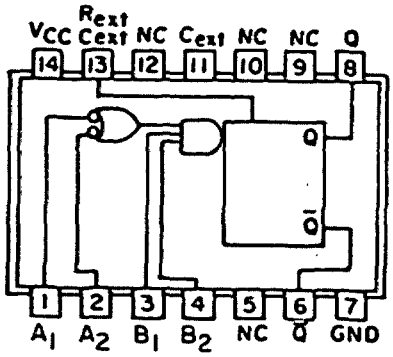
NOTE (device types 02 and 03):
An external timing capacitor may be connected between C_X and R_X/C_X (positive).

FIGURE 1. Logic diagrams and terminal connections (top view).

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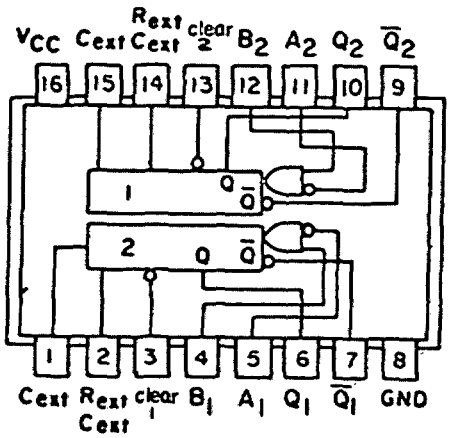
Device type 04

Cases A, B, C, and D



Device type 05

Cases E and F



NOTE (device types 04 and 05):
An external timing capacitor (C_X) and an external timing resistor (R_X) are required between Cext and R_{ext}/C_{ext} to determine output pulse duration and accuracy.

FIGURE 1. Logic diagrams and terminal connections (top view) - Continued.

Device type 01

INPUTS			OUTPUTS	
A_1	A_2	B	Q	$\overline{Q}$
H	H	X	L	H
X	L	↓	L	H
L	X	↓	L	H
L	X	↑	⎍	⎎
X	L	↑	⎍	⎎
H	↓	H	⎍	⎎
↓	H	H	⎍	⎎
↓	↓	H	⎍	⎎
X	↑	L	L	H
↑	X	L	L	H
H	↑	H	L	H
↑	H	H	L	H
↑	↑	H	L	H

Device type 02

INPUTS					OUTPUTS	
CLEAR	A_1	A_2	B_1	B_2	Q	$\overline{Q}$
L	X	X	X	X	L	H
X	H	H	X	X	L	H
X	X	X	L	X	L	H
X	X	X	X	L	L	H
X	L	X	H	H	L	H
H	L	X	↓	H	⎍	⎎
H	L	X	H	↓	⎍	⎎
H	X	L	H	H	L	H
H	X	L	↓	H	⎍	⎎
H	X	L	H	↓	⎍	⎎
H	H	↓	H	H	⎍	⎎
H	↓	↓	H	H	⎍	⎎
H	↓	H	H	H	⎍	⎎

Device type 03

INPUTS			OUTPUTS	
CLEAR	A	B	Q	$\overline{Q}$
L	X	X	L	H
X	H	X	L	H
X	X	L	L	H
H	L	↓	⎍	⎎
H	↓	H	⎍	⎎
↓	L	H	⎍	⎎

NOTE:
H = high level (steady state), L = low level (steady state), ↓ = transition from low to high level, ↓ = transition from high to low level, ⎍ = one high-level pulse, ⎎ = one low-level pulse, X = irrelevant (any input, including transitions).

FIGURE 2. Truth tables.

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Device type 04

INPUTS				OUTPUTS	
A_1	A_2	B_1	B_2	Q	$\overline{Q}$
H	H	X	X	L	H
X	X	L	X	L	H
X	X	X	L	L	H
L	X	H	H	L	H
L	X	↓	H	⎍	⎌
L	X	H	↓	⎍	⎌
X	L	H	H	L	H
X	L	↓	H	⎍	⎌
X	L	H	↓	⎍	⎌
H	↓	H	H	⎍	⎌
↓	↓	H	H	⎍	⎌
↓	H	H	H	⎍	⎌

Device type 05

INPUTS			OUTPUTS	
CLEAR	A	B	Q	$\overline{Q}$
L	X	X	L	H
H	H	↓	⎍	⎌
H	↓	L	⎍	⎌

NOTE:
H = high level (steady state), L = low level (steady state), ↓ = transition from low to high level, ↓ = transition from high to low level, ⎍ = one high-level pulse, ⎌ = one low-level pulse, x = irrelevant (any input, including transitions).

FIGURE 2. Truth tables - Continued.

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Description of device types 01 and 02

These monostables are designed to provide the system designer with complete flexibility in controlling the pulse width, either to lengthen the pulse by retriggering, or to shorten by clearing. Device types 01 and 02 have an internal timing resistor which allows the circuit to be operated with only an external capacitor, if so desired. Applications requiring more precise pulse widths and not requiring the clear feature can best be satisfied with device type 01.

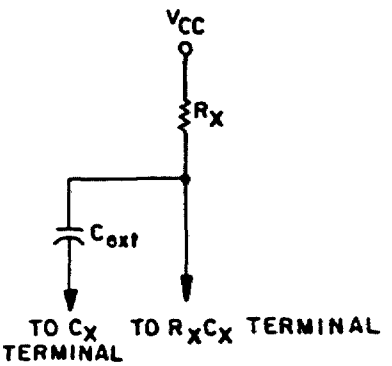
The output pulse is primarily a function of the external capacitor and resistor. For $C_X >$ 1,000 pF, the output pulse width (t_w) is defined as:

$$t_w \text{ (device type 01)} = R_X C_X \text{ Ln2}$$

$$t_w \text{ (device type 02)} = 0.32 R_X C_X \left(1 + \frac{0.7}{R_X}\right)$$

where R_X is in kΩ (either internal or external timing resistor)
C_X is in pF
t_w is in ns

For pulse widths when $C_X \leq 1,000$ pF, the following circuit for timing component connections is recommended.

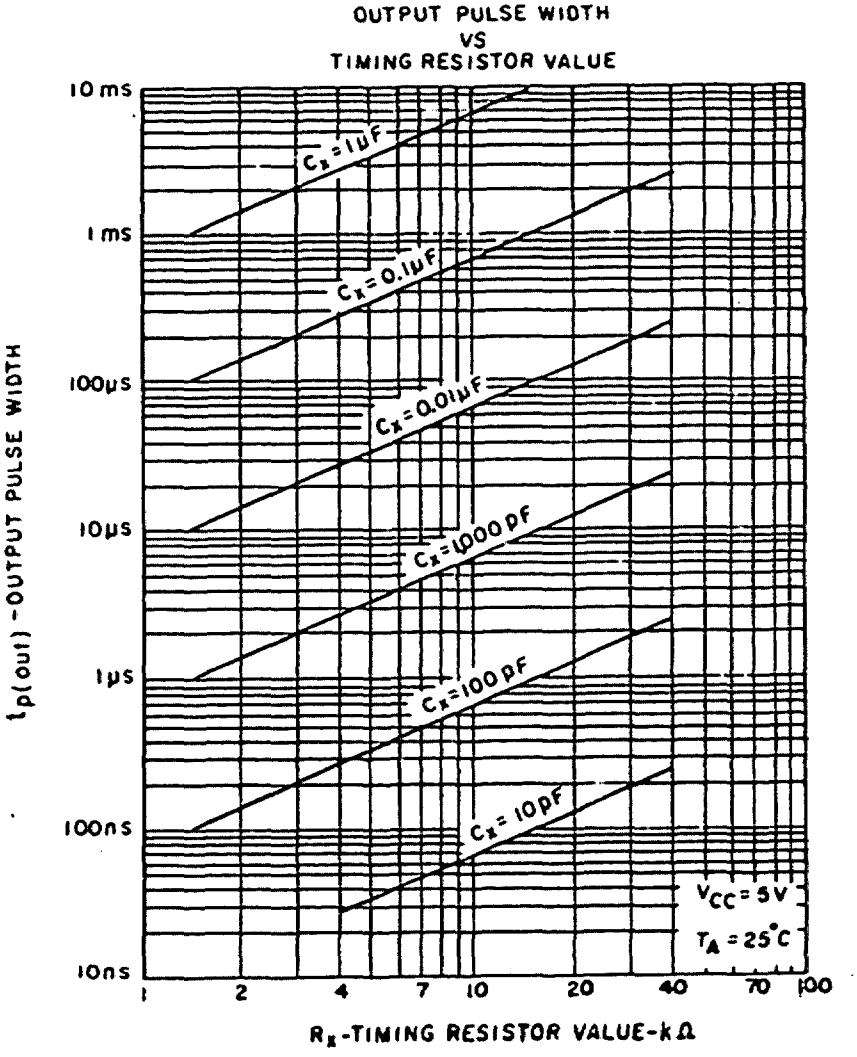


These circuits are fully compatible with most TTL or DTL families. Inputs are diode-clamped to minimize reflections due to transmission-line effects, which simplifies design. Typical power dissipation per one shot is 115 milliwatts; typical average propagation delay time to the Q output is 21 nanoseconds.

FIGURE 3. Device descriptions.

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Description of device type 01 - Continued



NOTE: See 1.4 for maximum external timing resistance values.

FIGURE 3. Device descriptions - Continued.

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8. Description of device type 01 - Continued.

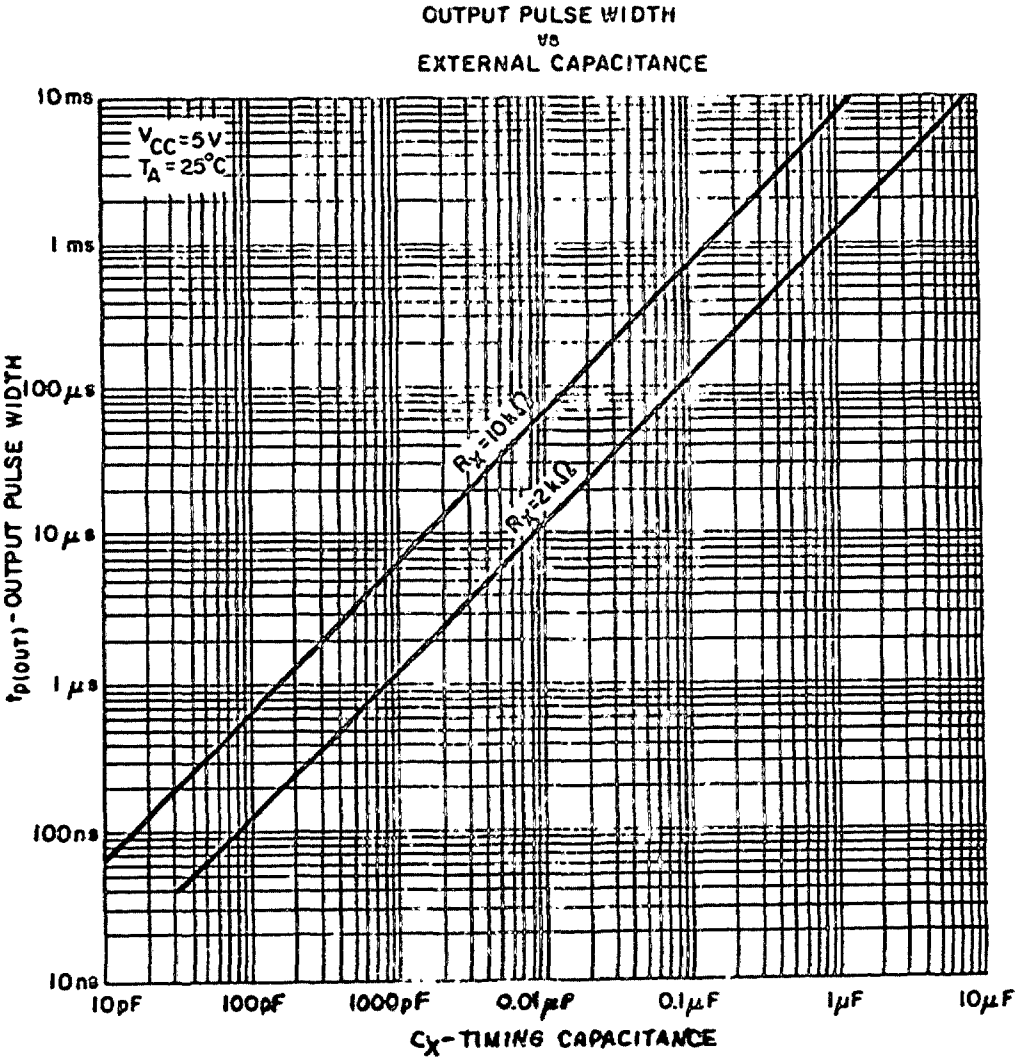
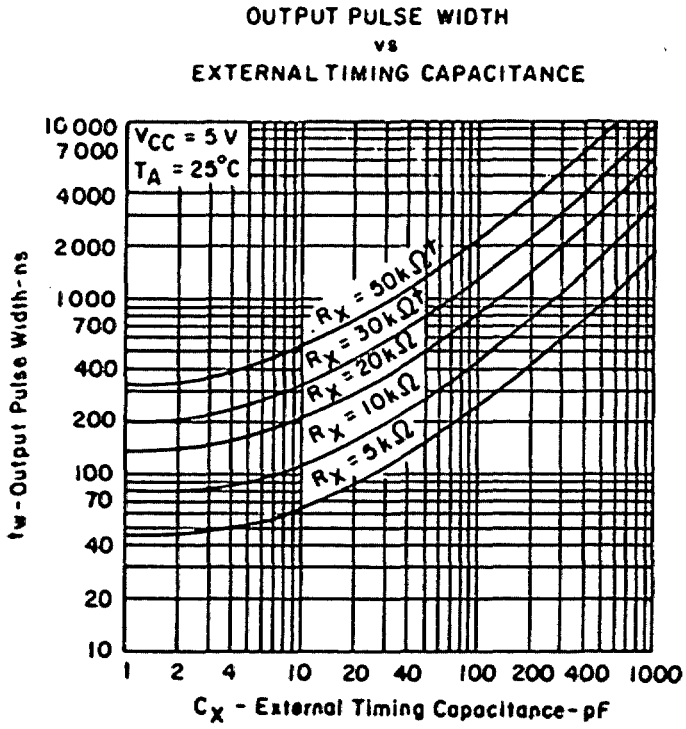


FIGURE 3. Device descriptions - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:50Z
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Description of device types 02 and 03



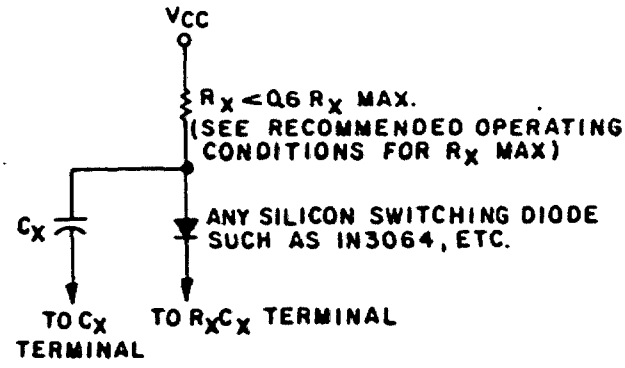
†These values of resistance exceed the maximums recommended for use over the full temperature range.

FIGURE 3. Device descriptions - Continued.

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Description of device types 02 and 03

To prevent reverse voltage across C_X, it is recommended that the following circuit be employed when using electrolytic capacitors and in applications utilizing the clear functions. This circuit is also recommended for $C_X > 1,000$ pF.



In all applications using the diode, the pulse width is:

$$t_w = 0.28\ R_X\ C_X \left(1 + \frac{0.7}{R_X}\right)$$

where R_X is in kΩ
C_X is in pF
t_w is in ns

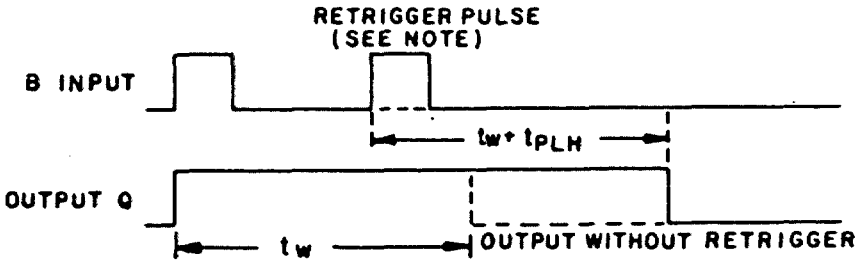
FIGURE 3. Device descriptions - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:50Z
Check the source to verify that this is the current version before use.

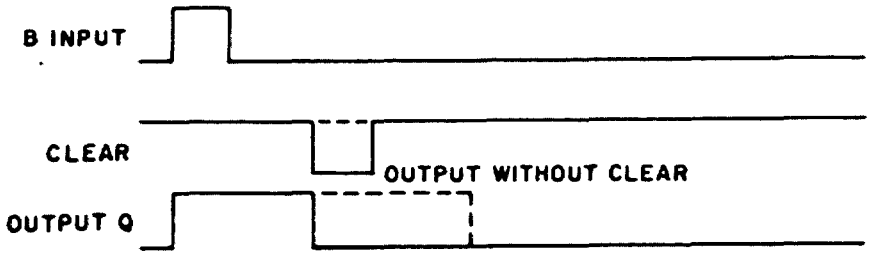
Description of device types 02 and 03

These monolithic TTL retriggerable monostable multivibrators feature d-c triggering from gated low-level-active (A) and high-level-active (B) inputs, and also provide overriding direct clear inputs. Complementary outputs are provided. The retrigger capability simplifies the generation of output pulses of extremely long duration. By triggering the input before the output pulse is terminated, the output pulse may be extended. The overriding clear capability permits any output pulse to be terminated at a predetermined time independently of the timing components R and C.

Waveforms below illustrates triggering the one-shot with the high-level-active (B) inputs.



OUTPUT PULSE CONTROL USING RETRIGGER PULSE



OUTPUT PULSE CONTROL USING CLEAR INPUT

TYPICAL INPUT/OUTPUT PULSES

NOTE: Retrigger pulse must not start before 0.22 C_X (in picofarads) nanoseconds after previous trigger pulse.

FIGURE 3. Device descriptions - Continued.

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Description of device type 04

This retriggerable monostable multivibrator provides an output pulse whose duration and accuracy is a function of external timing components. It is designed to allow a choice of triggering either the leading or trailing edge of the pulse, thus providing the system designer with complete flexibility in controlling the pulse width.

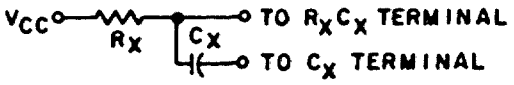
The output pulse width is primarily a function of the external capacitor and external resistor. For $C_X \geq 1,000$ pF, the output pulse width (t_w) is defined as:

$$t_w = 0.32\ R_X\ C_X\ \left(1 + \frac{0.7}{R_X}\right)$$

where R_X is in kΩ (see note 3)
C_X is in pF (see note 2)
t_w is in ns

For pulse widths when $C_X < 1,000$ pF, the following circuits for timing component connections are recommended.

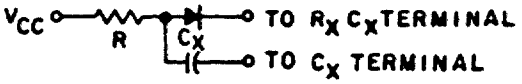
Circuit A. This circuit is for use with low leakage electrolytic capacitors. This configuration can be used predictably only if the forward capacitor leakage at 5.0 volts is less than 3 μA, and the reverse capacitor leakage at 1.0 volt is less than 5 μA over the operational temperature range, and note 1 below is satisfied.



Circuit B. This circuit is for use with high reverse leakage current electrolytic capacitors. The diode in this configuration prevents high reverse leakage currents through the capacitor by preventing a reverse voltage across the capacitor.

$t_w \approx 0.3\ RC_X$

Any silicon switching diode such as 1N3064, etc.



$R < 0.6\ R_X$ (max.) (see recommended operating conditions for R_X max.)

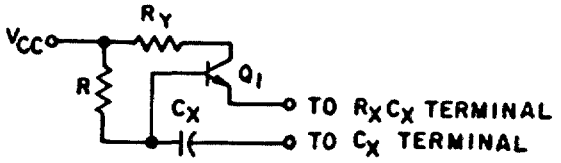
FIGURE 3. Device descriptions - Continued.

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Description of device type 04 - Continued

Circuit C. This circuit is used to obtain extended pulse widths. This configuration obtains extended pulse widths, because of the larger timing resistor allowed by Beta multiplication. Electrolytics with high (>5 μA) reverse leakage currents can be used.

$R < R_X\ (0.7)\ (h_{FE}{}^{Q1})$ or $< 2.5\ M\Omega$ whichever is lesser.
R_X (min) $< R_Y < R_X$ (max) ($5 \leq R_Y \leq 10\ k\Omega$ is recommended.
Q1: NPN silicon transistor with h_{FE} requirements of above equations, such as 2N5961 or 2N5962.
$t_w \approx 0.3\ RC_X$



NOTES:
1. C_X may vary from 0 to any necessary value available. If however, the capacitor has leakage approaching 3.0 μA or if stray capacitance from either terminal to ground is more than 50 pF, the timing equations may not represent the pulse width obtained.
2. Configurations B and C are not recommended with retriggerable operation.
3. R_X may vary from 5.0 to 25 kΩ.

FIGURE 3. Device descriptions - Continued.

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Description of device type 04

OUTPUT PULSE WIDTH
vs
TIMING RESISTANCE AND CAPACITANCE

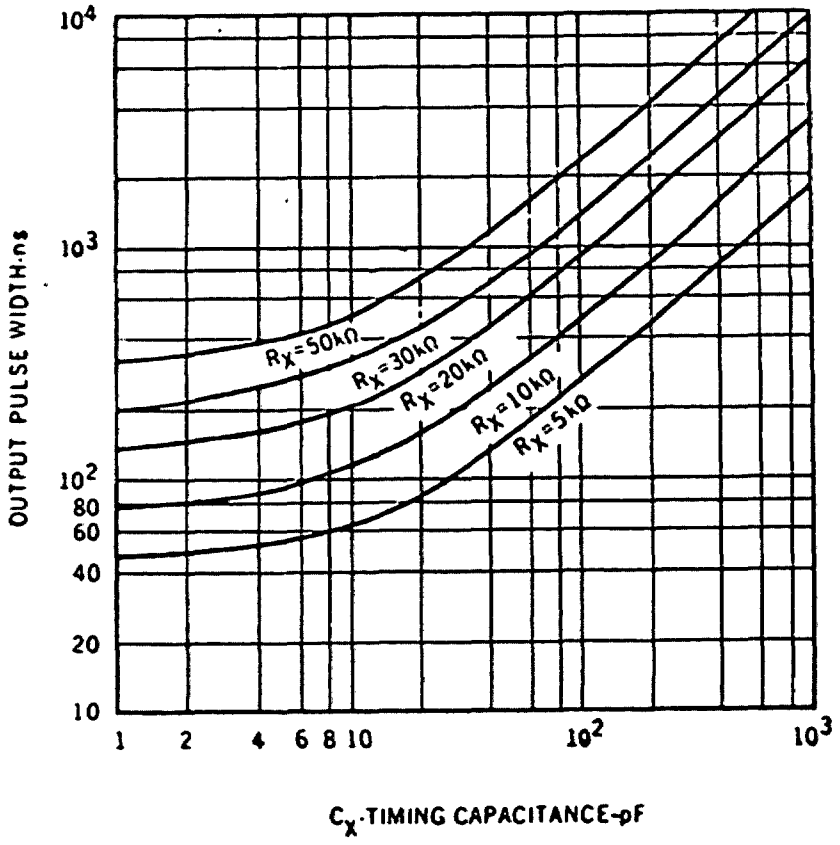


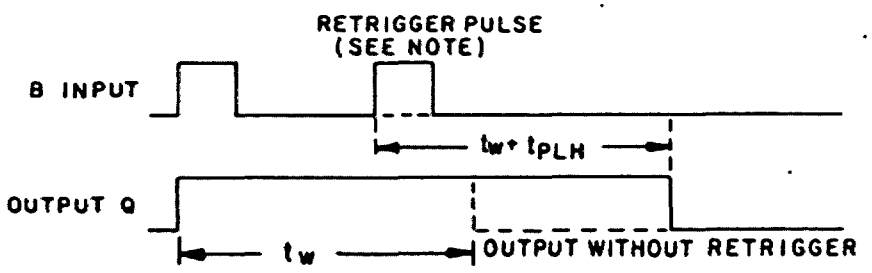
FIGURE 3. Device descriptions - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:50Z
Check the source to verify that this is the current version before use.

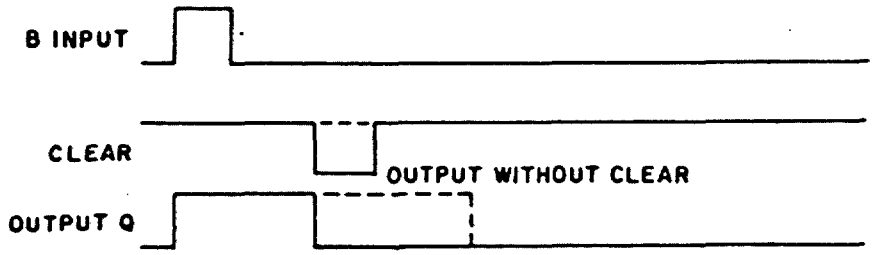
Description of device types 04 and 05

These monolithic TTL retriggerable monostable multivibrators feature d-c triggering from gated low-level-active (A) and high-level-active (B) inputs, and also provide overriding direct clear inputs. Complementary outputs are provided. The retrigger capability simplifies the generation of output pulses of extremely long duration. By triggering the input before the output pulse is terminated, the output pulse may be extended. For device type 05, the overriding clear permits any output pulse to be terminated at a predetermined time independently of the timing components R and C.

Waveforms below illustrates triggering the one-shot with the high-level-active (B) inputs.



OUTPUT PULSE CONTROL USING RETRIGGER PULSE



OUTPUT PULSE CONTROL USING CLEAR INPUT

TYPICAL INPUT/OUTPUT PULSES

NOTE: Retrigger pulse must not start before $\approx 0.3\ C_X$ nanoseconds after previous trigger pulse.

FIGURE 3. Device descriptions - Continued.

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Description of device type 05

This dual retriggerable, resettable monostable multivibrator provides an output pulse whose duration and accuracy is a function of external timing components. It is designed to allow a choice of triggering either the leading edge or trailing edge of the pulse, thus providing the system designer with complete flexibility in controlling the pulse width.

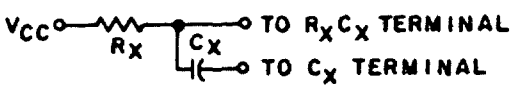
The output pulse width is primarily a function of the external capacitor and external resistor. For $C_X \geq 1,000$ pF, the output pulse width (t_w) is defined as:

$$t_w = 0.31\ R_X\ C_X \left(1 + \frac{1}{R_X}\right)$$

where R_X is in kΩ (see note 3)
C_X is in pF (see note 2)
t_w is in ns

For pulse widths when $C_X < 1,000$ pF, the following circuits for timing component connections are recommended.

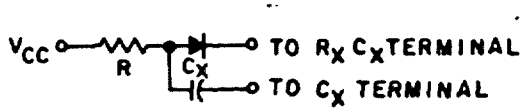
Circuit A. This circuit is for use with low leakage electrolytic capacitors. This configuration can be used predictably only if the forward capacitor leakage at 5.0 volts is less than 3 μA, and the reverse capacitor leakage at 1.0 volt is less than 5 μA over the operational temperature range, and note 1 below is satisfied.



Circuit B. This circuit is for use with high reverse leakage current electrolytic capacitors. The diode in this configuration prevents high reverse leakage currents through the capacitor by preventing a reverse voltage across the capacitor.

$$t_w \approx 0.3\ RC_X$$

Any silicon switching diode such as 1N3064, etc.



$R < 0.6\ R_X$ (max.) (see recommended operating conditions for R_X max.)

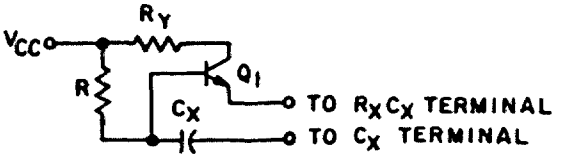
FIGURE 3. Device descriptions - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:50Z
Check the source to verify that this is the current version before use.

Description of device type 05 - Continued

Circuit C. This circuit is used to obtain extended pulse widths. This configuration obtains extended pulse widths, because of the larger timing resistor allowed by Beta multiplication. Electrolytics with high inverse leakage currents can be used.

$R < R_X\ (0.7)\ (h_{FE}Q1)$ or $< 2.5\ M\Omega$ whichever is the lesser.
$R_X\ (min) < R_Y < R_X\ (max)$ ($5 \leq R_Y \leq 10\ k\Omega$ is recommended).
Q1: NPN silicon transistor with h_{FE} requirements of above equations, such as 2N5961 or 2N5962.
$t_w \approx 0.3\ RC_X$



NOTES:
1. C_X may vary from 0 to any necessary value available. If however, the capacitor has leakage approaching 3.0 µA or if stray capacitance from either terminal to ground is more than 50 pF, the timing equations may not represent the pulse width obtained.
2. Configurations B and C are not recommended with retriggerable operation.
3. R_X may vary from 5.0 to 25 kΩ.

FIGURE 3. Device descriptions - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:50Z
Check the source to verify that this is the current version before use.

Description of device type 05

OUTPUT PULSE WIDTH
vs
TIMING RESISTANCE AND CAPACITANCE

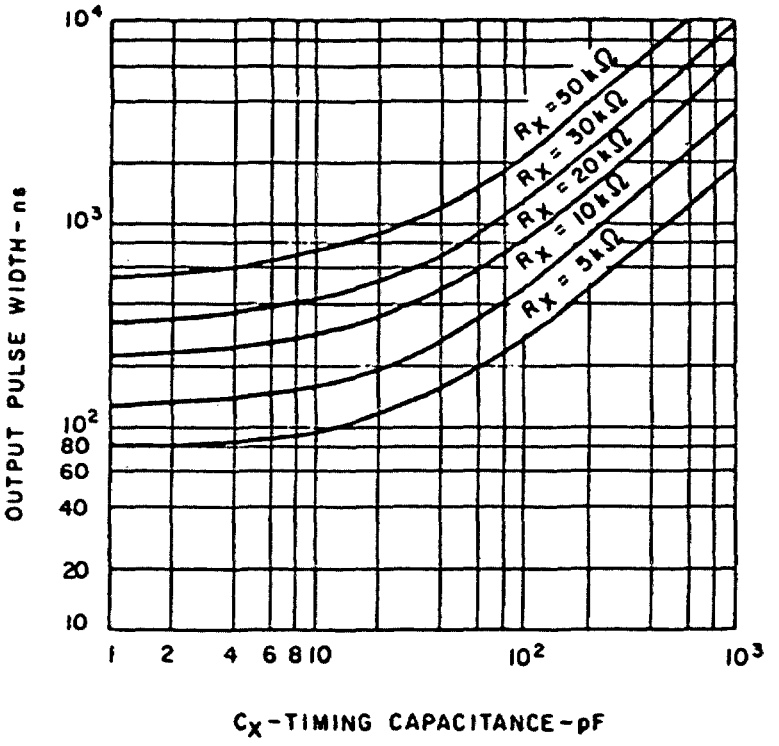
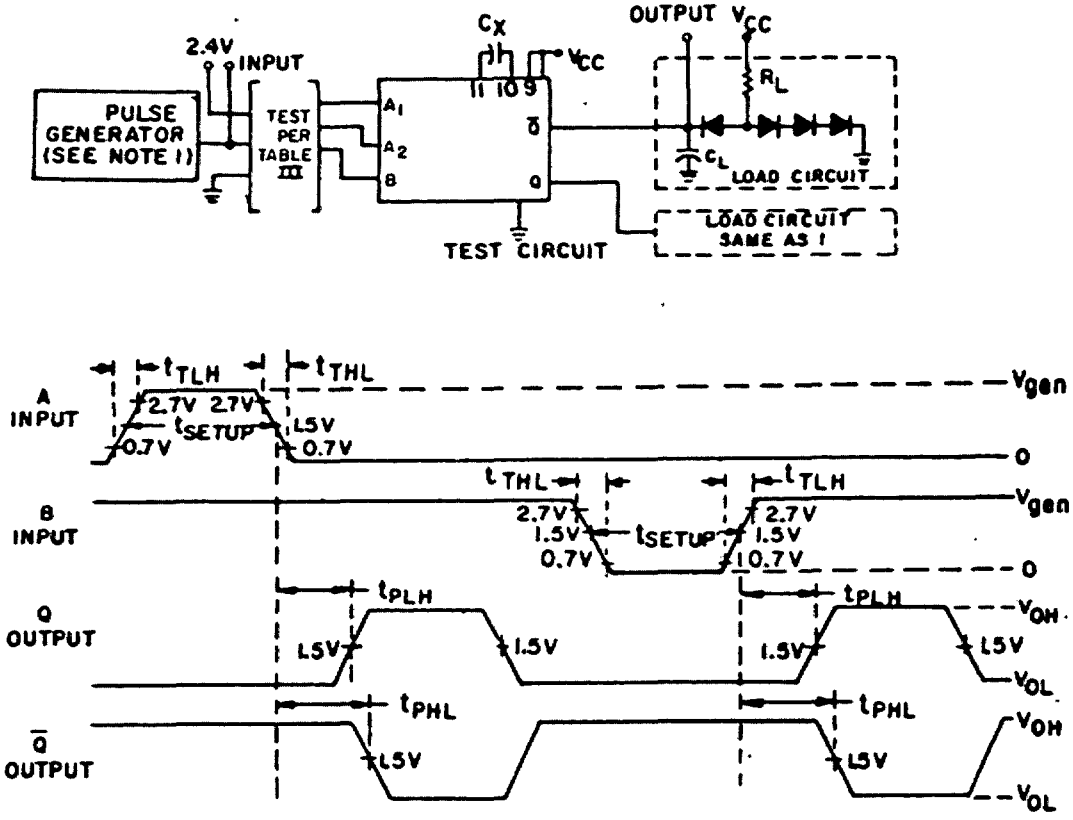


FIGURE 3. Device descriptions - Continued.

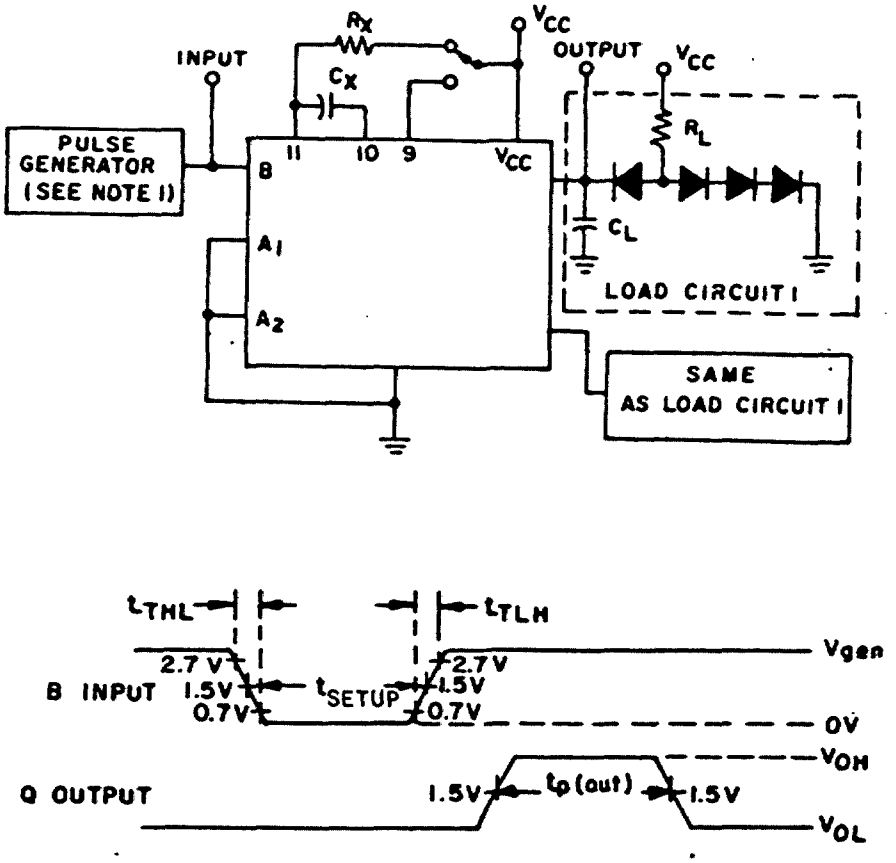
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:50Z
Check the source to verify that this is the current version before use.



NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3.0 V, $t_{THL} \leq$ 10 ns, $t_{TLH} \leq$ 10 ns, t_{SETUP} = 60 ns, PRR ≤ 1 MHz, and $Z_{out} \approx 50\Omega$.
2. All diodes are 1N3064 or equivalent.
3. C_L = 50 pF minimum including probe and jig capacitance.
4. R_L = 390Ω ±5%.
5. V_{CC} = 5.0 V minimum.
6. See table III for R_X and C_X values.

FIGURE 4. Switching test circuit for t_{PHL} and t_{PLH} of device type 01.

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Check the source to verify that this is the current version before use.



NOTES:

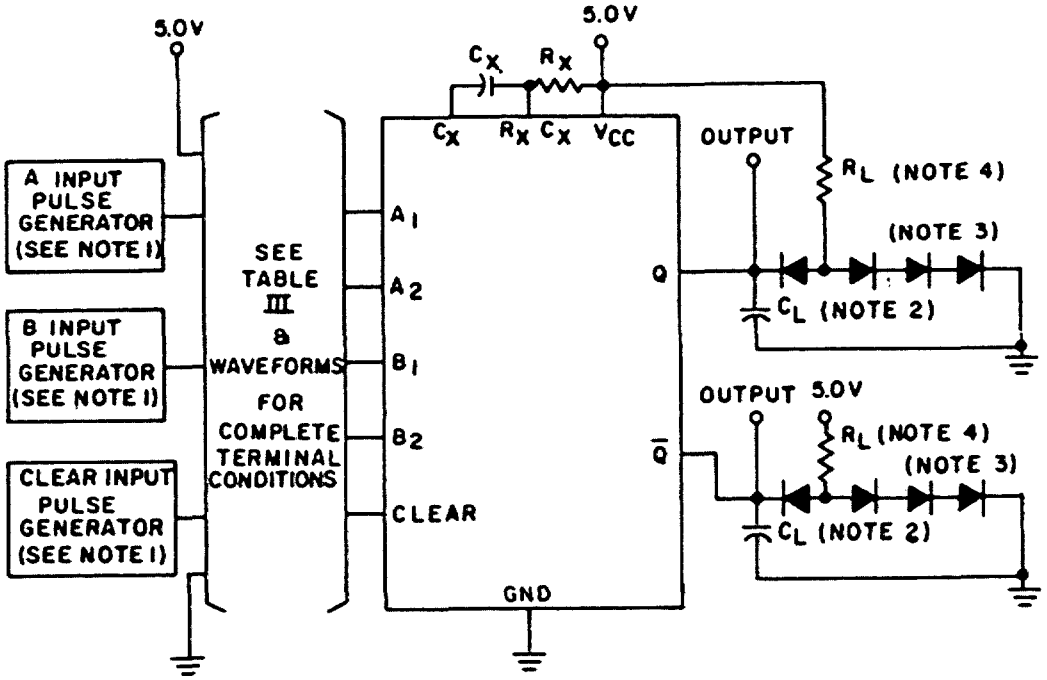
1. The pulse generator has the following characteristics: V_{gen} = 3.0 V, $t_{THL} \leq 10$ ns, $t_{TLH} \leq 10$ ns, t_{SETUP} = 60 ns, $Z_{out} \approx 50\Omega$ and PRR is as follows:

TEST	PRR
t_p(out)1 and t_p(out)2	1 MHz
t_p(out) 3	500 kHz
t_p(out) 4	20 KHz

2. V_{CC} = 5.0 V minimum; R_L = 390Ω ±5%; C_L = 50 pF minimum including jig and probe capacitance.
3. See table III for R_X and C_X values.
4. All diodes are 1N3064 or equivalent.

FIGURE 5. Test circuit for t_{SETUP} and t_p(out) of device type 01.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:50Z
Check the source to verify that this is the current version before use.

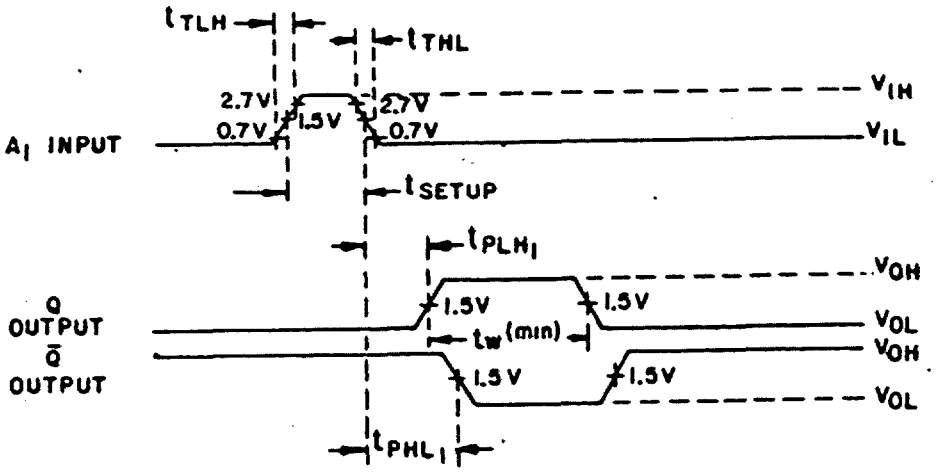


NOTES:
1. Unless otherwise specified in the notes with the individual waveforms, all pulse generators shall have the following characteristics: $t_{TLH} \leq 10$ ns, $t_{THL} \leq 10$ ns, $V_{IH} = 3.0$ V minimum, $V_{IL} = 0$ V, $Z_{out} \approx 50\Omega$.
2. $C_L = 50$ pF minimum, including jig and probe capacitance.
3. All diodes are 1N3064,or equivalent.
4. $R_L = 390\Omega \pm 5\%$.
5. See table III for R_X and C_X values.

FIGURE 6. Switching test circuit and waveforms for device type 02.

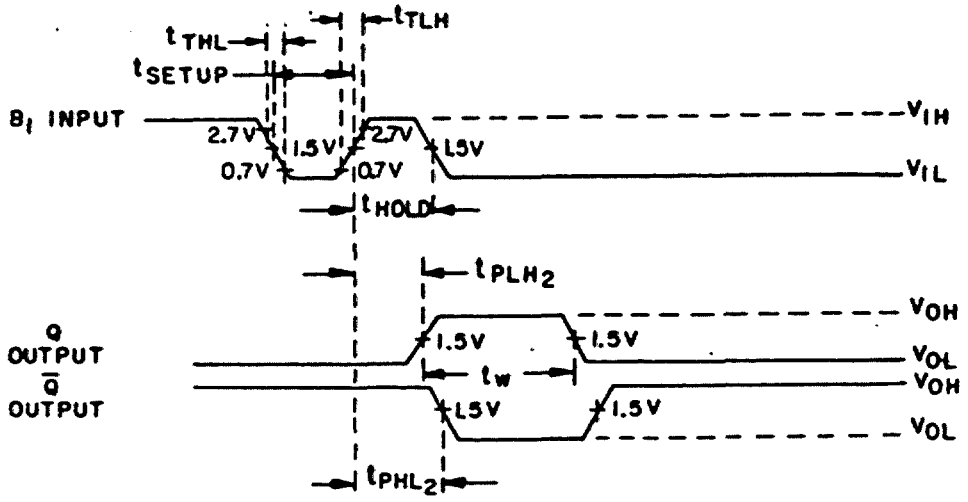
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:50Z
Check the source to verify that this is the current version before use.

A_1 INPUT to Q and $\overline{Q}$ OUTPUTS (t_{PLH1}, t_{PHL1}) (t_W (min))



NOTES:
1. A_1 input characteristics: PRR ≤ 1 MHz, t_{SETUP} = 40 ns.
2. A_2, B_1, B_2, and clear = 5.0 V.

B1 INPUT to Q and $\overline{Q}$ OUTPUTS (t_{PLH2}, t_{PHL2}) (t_W)

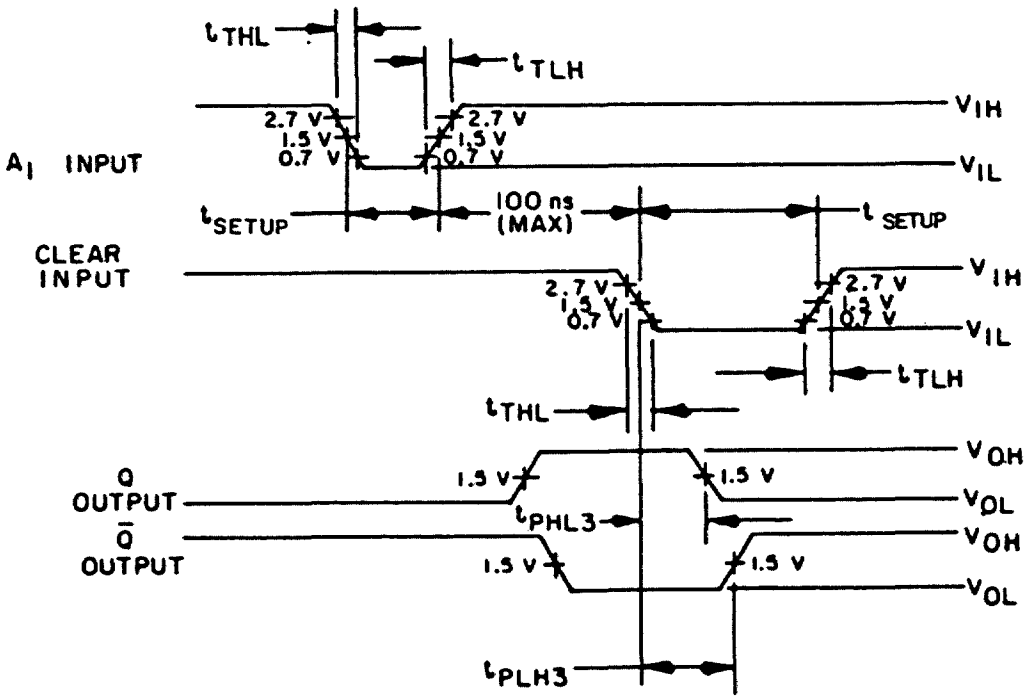


NOTES:
1. B_1 input characteristics: PRR ≤ 1 MHz, t_{SETUP} = 40 ns, t_{HOLD} = 40 ns.
2. A_1, A_2 = GND, B_2, clear = 5.0 V.

FIGURE 6. Switching test circuit and waveforms for device type 02 - Continued.

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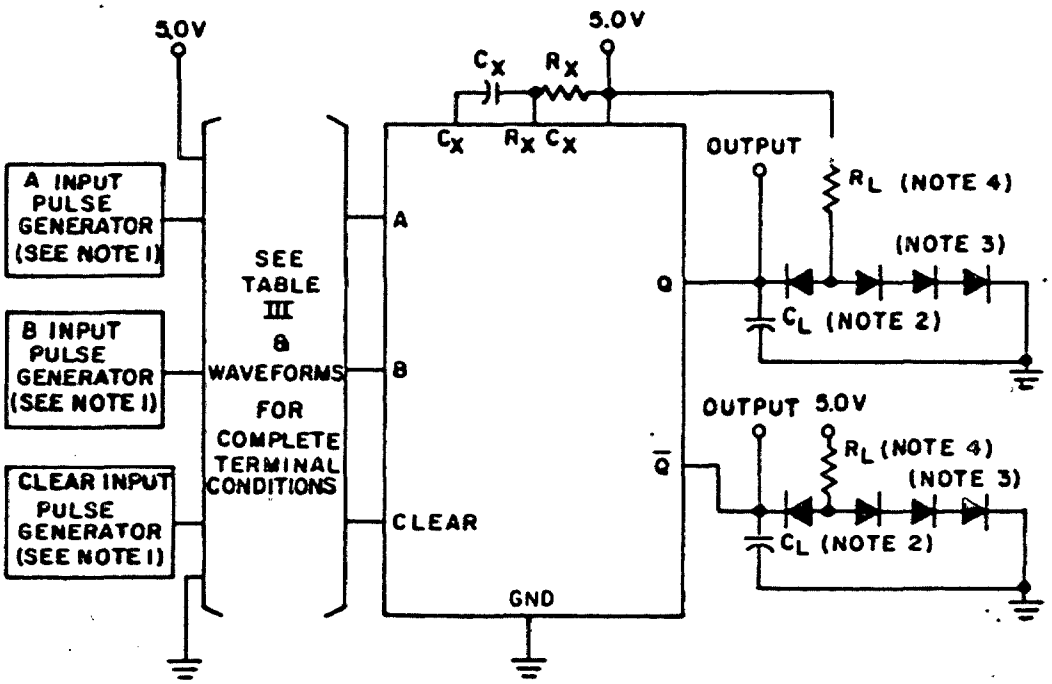
CLEAR to Q and $\overline{Q}$ OUTPUTS (t_{PLH3} & t_{PHL3})



NOTES:
1. A_1 input characteristics: PRR $\leq$ 1 MHz, t_{SETUP} = 40 ns.
2. Clear input characteristics: PRR $\leq$ 1 MHz, t_{SETUP} = 100 ns.
3. A_2, B_1, B_2 = 5.0 V.

FIGURE 6. Switching test circuit and waveforms for device type 02 - Continued.

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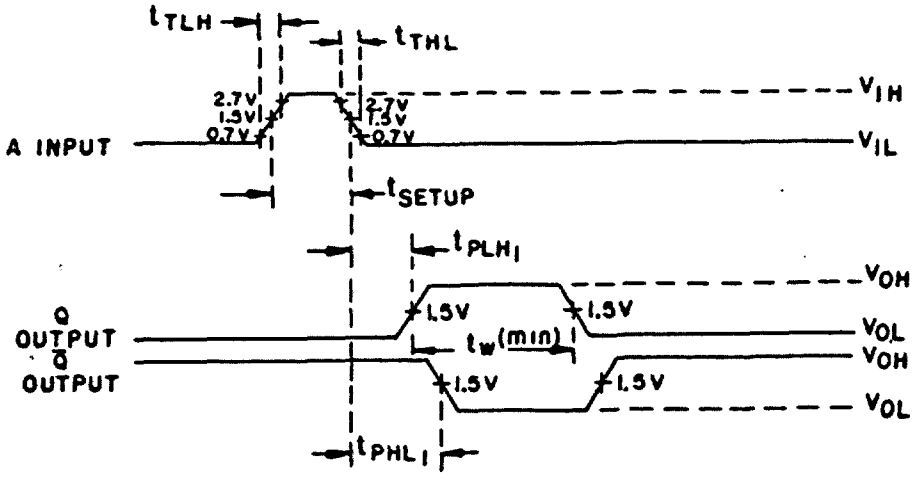


NOTES:
1. Unless otherwise specified in the notes with the individual waveforms, all pulse generators shall have the following characteristics: $t_{TLH} \leq 10$ ns, $t_{THL} \leq 10$ ns, $V_{IH} = 3.0$ V minimum, $V_{IL} = 0$ V, $Z_{out} \approx 50\Omega$.
2. $C_L = 50$ pF minimum, including jig and probe capacitance.
3. All diodes are 1N3064 or equivalent.
4. $R_L = 390\Omega \pm 5\%$.
5. See table III for R_X and C_X values.

FIGURE 7. Switching test circuit and waveforms for device type 03.

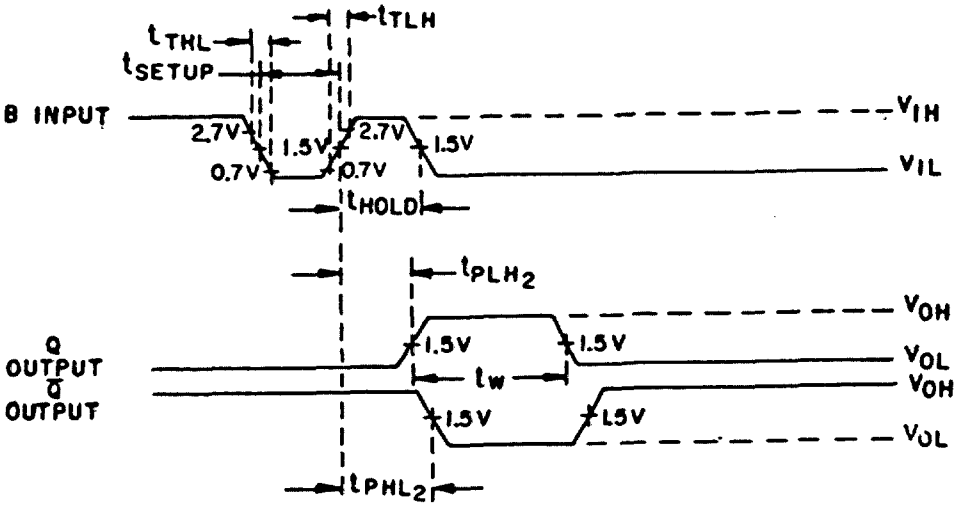
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:50Z
Check the source to verify that this is the current version before use.

A INPUT to Q and $\overline{Q}$ OUTPUTS (t_{PLH1}, t_{PHL1}) (t_W (min))



NOTES:
1. A input characteristics: PRR = 1 MHz, t_{SETUP} = 40 ns.
2. B, clear = 5.0 V.

B INPUT to Q and $\overline{Q}$ OUTPUTS (t_{PLH2}, t_{PHL2}) (t_W)

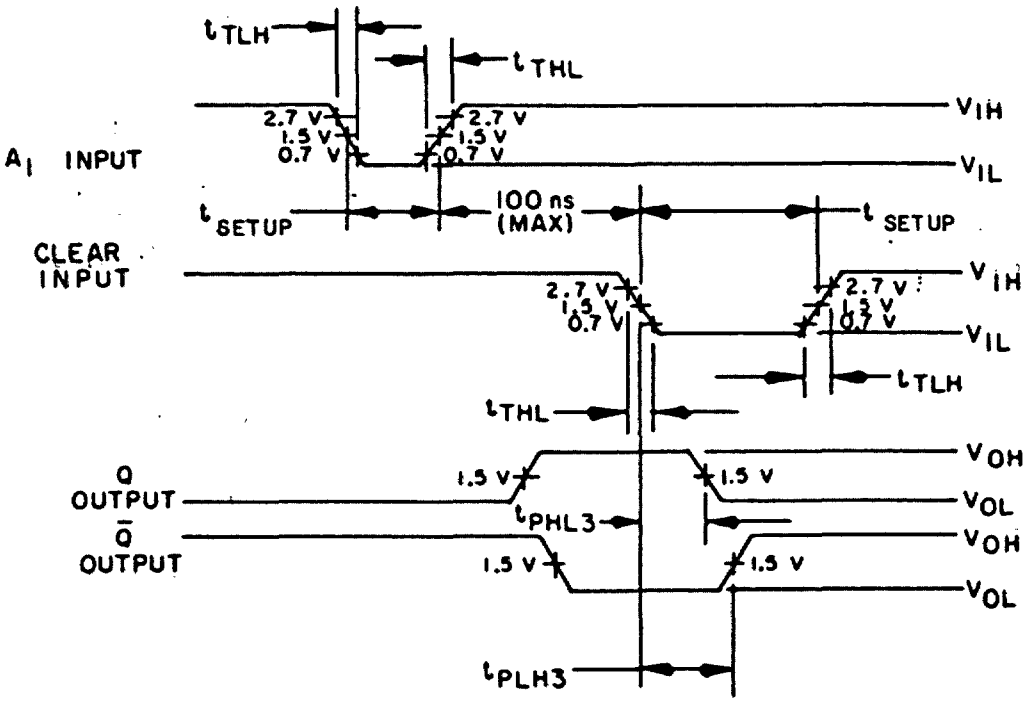


NOTES:
1. B input characteristics: PRR $\leq$ 285 kHz, t_{SETUP} = 40 ns, t_{HOLD} = 40 ns.
2. A = GND, clear = 5.0 V.

FIGURE 7. Switching test circuit and waveforms for device type 03 - Continued.

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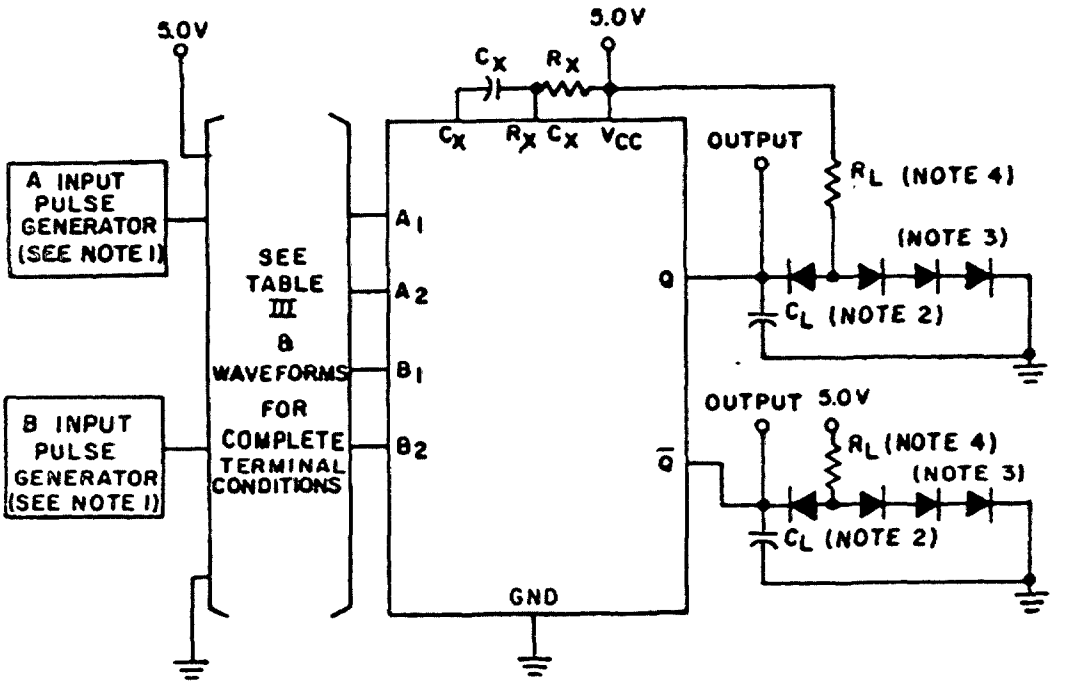
CLEAR to Q and $\overline{Q}$ OUTPUTS (t_{PLH3} & t_{PHL3})



NOTES:
1. A_1 input characteristics: PRR $\leq$ 285 kHz, t_{SETUP} = 40 ns.
2. Clear input characteristics: PRR $\leq$ 285 kHz, t_{SETUP} = 100 ns.
3. B = 5.0 V.

FIGURE 7. Switching test circuit and waveforms for device type 03 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:50Z
Check the source to verify that this is the current version before use.

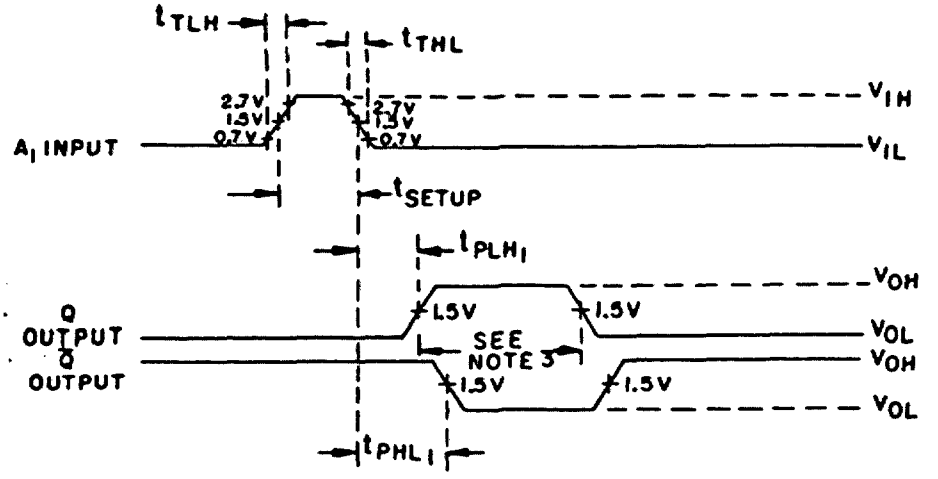


NOTES:
1. Unless otherwise specified in the notes with the individual waveforms, all pulse generators shall have the following characteristics: $t_{TLH} \leq 10$ ns, $t_{THL} \leq 10$ ns, $V_{IH} = 3.0$ V minimum, $V_{IL} = 0$ V, $Z_{out} \approx 50\Omega$.
2. $C_L = 50$ pF minimum, including jig and probe capacitance.
3. All diodes are 1N3064 or equivalent.
4. $R_L = 390\Omega \pm 5\%$.
5. See table III for R_X and C_X values.

FIGURE 8. Switching test circuit and waveforms for device type 04.

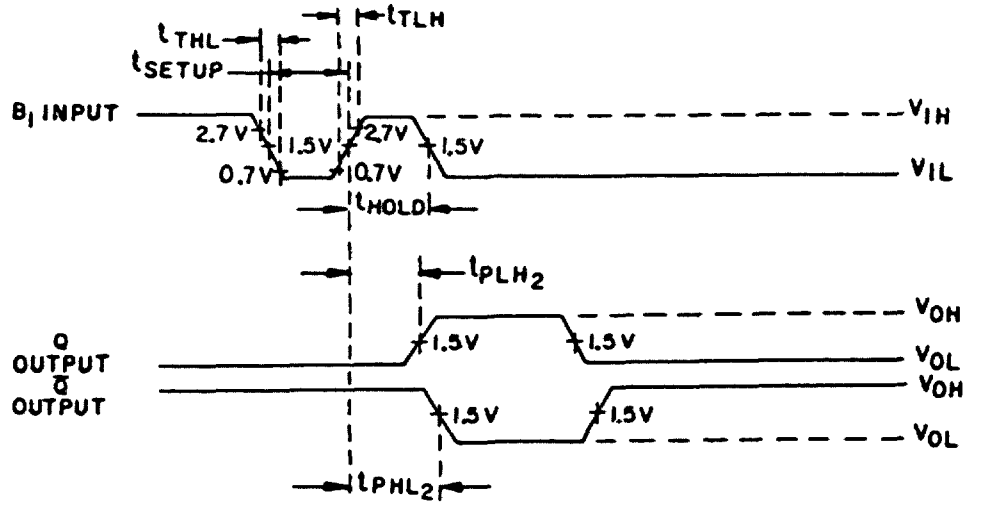
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:50Z
Check the source to verify that this is the current version before use.

A_1 INPUT to Q and $\overline{Q}$ OUTPUTS (t_{PLH1}, t_{PHL1}) (t_W) (t_W (min))



NOTES:
1. A_1 input characteristics: PRR $\leq$ 1 MHz, t_{SETUP} = 40 ns.
2. A_2, B_1 and B_2 = 5.0 V.
3. t_W and t_W (min).

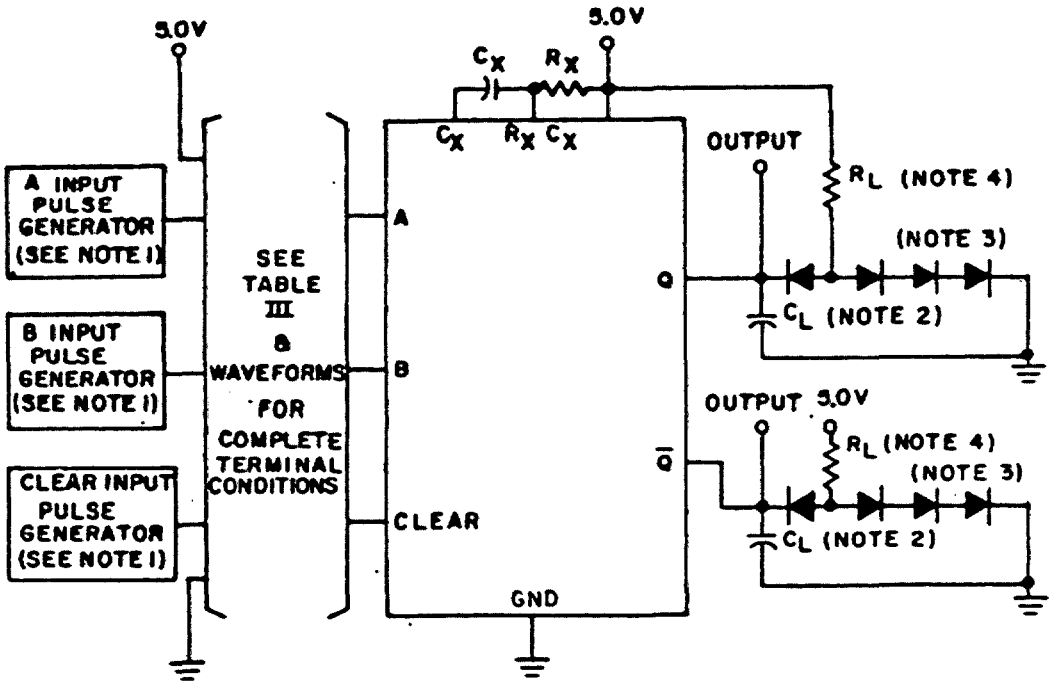
B_1 INPUT to Q and $\overline{Q}$ OUTPUTS (t_{PLH2}, t_{PHL2})



NOTES:
1. B_1 input characteristics: PRR $\leq$ 1 MHz, t_{SETUP} = 40 ns, t_{HOLD} = 40 ns.
2. A_1, A_2 = GND, B_2 = 5.0 V.

FIGURE 8. Switching test circuit and waveforms for device type 04 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:50Z
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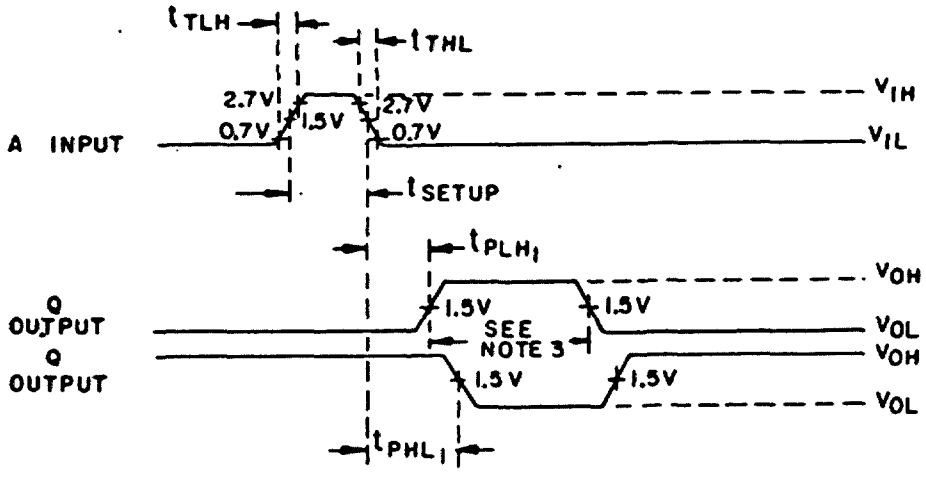


NOTES:
1. Unless otherwise specified in the notes with the individual waveforms, all pulse generators shall have the following characteristics: $t_{TLH} \leq 10$ ns, $t_{THL} \leq 10$ ns, $V_{IH} = 3.0$ V minimum, $V_{IL} = 0$ V, $Z_{out} \approx 50\Omega$.
2. $C_L = 50$ pF minimum, including jig and probe capacitance.
3. All diodes are 1N3064 or equivalent.
4. $R_L = 390\Omega \pm 5\%$.
5. See table III for R_X and C_X values.

FIGURE 9. Switching test circuit and waveforms for device type 05.

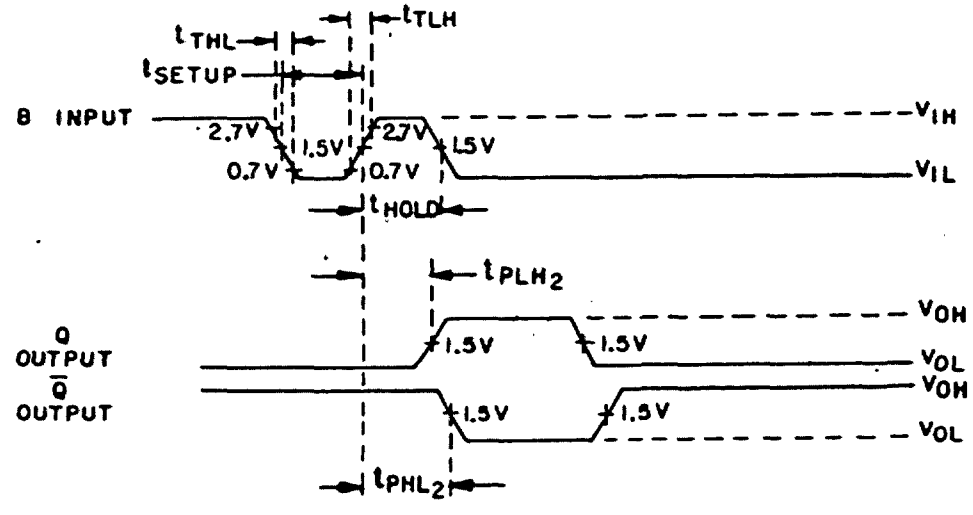
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:50Z
Check the source to verify that this is the current version before use.

A INPUT to Q and $\overline{Q}$ OUTPUTS (t_{PLH1}, t_{PHL1}) (t_W) (t_W (min))



NOTES:
1. A input characteristics: PRR ≤ 1 MHz, t_{SETUP} = 40 ns.
2. B = GND, clear = 5.0 V.
3. t_W and t_W (min).

B INPUT to Q and $\overline{Q}$ OUTPUTS (t_{PLH2}, t_{PHL2})

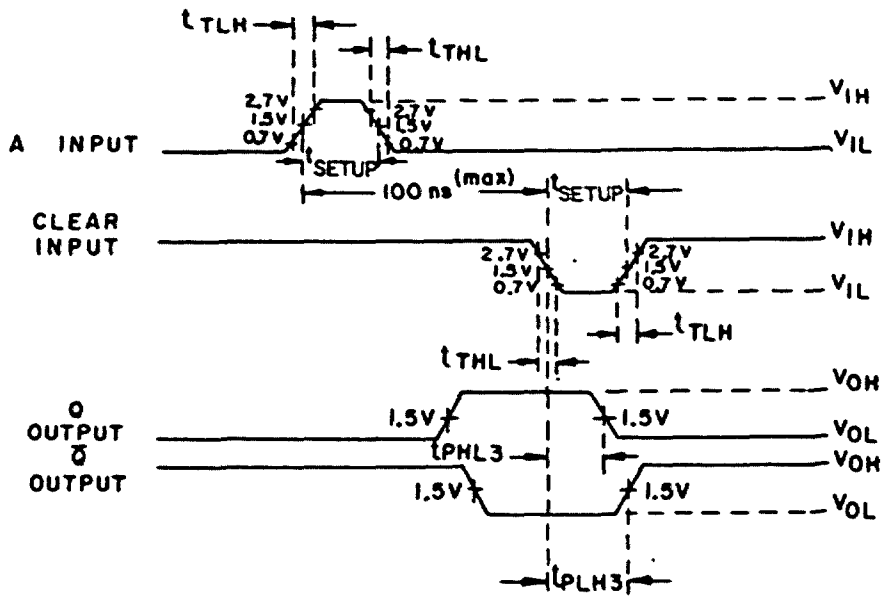


NOTES:
1. B input characteristics: PRR ≤ 1 MHz, t_{SETUP} = 40 ns, t_{HOLD} = 40 ns.
2. A and clear = 5.0 V.

FIGURE 9. Switching test circuit and waveforms for device type 05 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:50Z
Check the source to verify that this is the current version before use.

CLEAR to Q and $\overline{Q}$ OUTPUTS (t_{PLH3} & t_{PHL3})

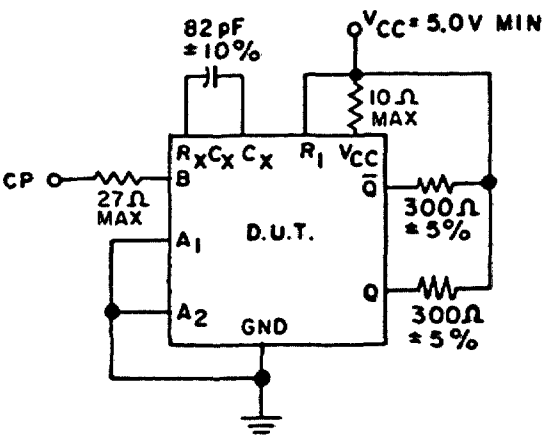


NOTES:
1. A input characteristics: PRR $\leq$ 1 MHz, t_{SETUP} = 40 ns.
2. Clear input characteristics: PRR $\leq$ 1 MHz, t_{SETUP} = 40 ns.
3. B = GND.

FIGURE 9. Switching test circuit and waveforms for device type 05 - Continued.

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Device type 01

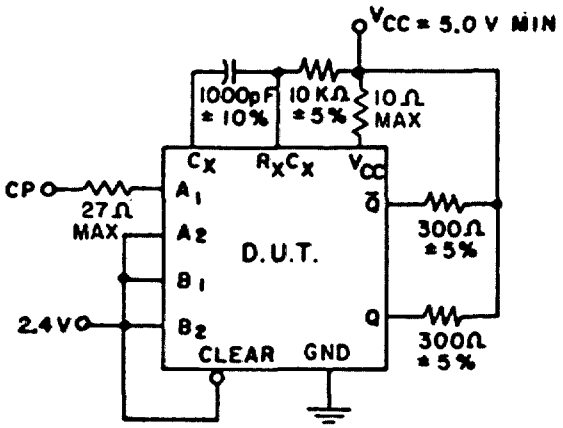


NOTE: CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%, V_{IH} = 2.0 V minimum to 5.5 V maximum; V_{IL} = -0.5 V minimum to 0.8 V maximum.

FIGURE 10. Burn-in and life test circuits.

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Device type 02

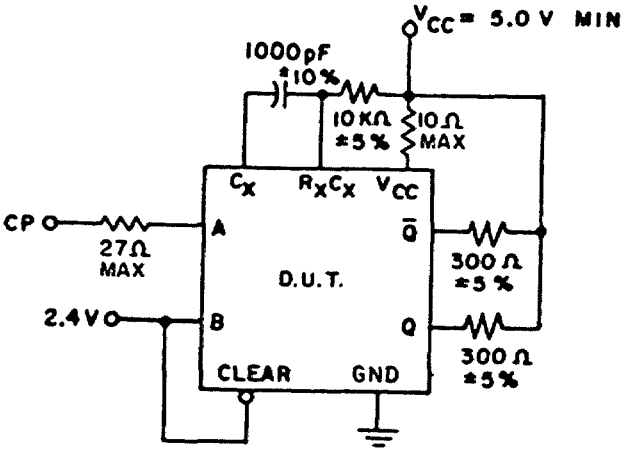


NOTE: CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%, V_{IH} = 2.0 V minimum to 5.5 V maximum; V_{IL} = -0.5 V minimum to 0.8 V maximum.

FIGURE 10. Burn-in and life test circuits - Continued.

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Device type 03

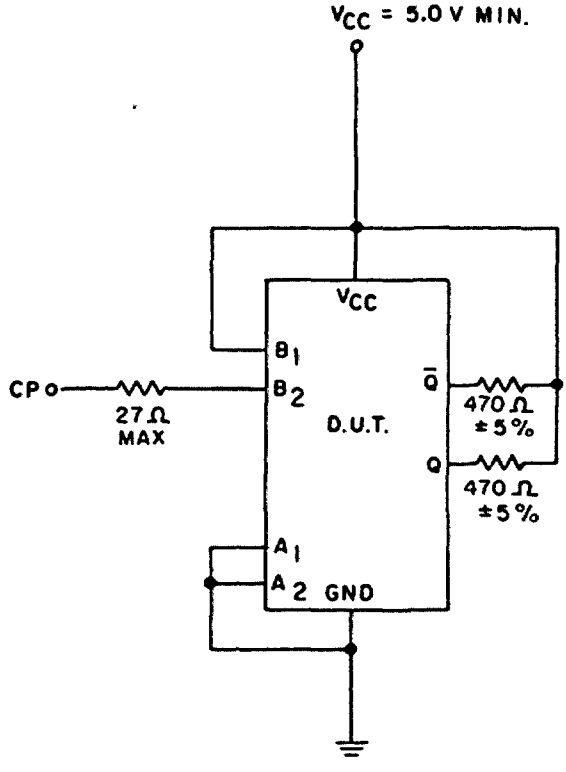


NOTES:
1. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%, V_{IH} = 2.0 V minimum to 5.5 V maximum; V_{IL} = -0.5 V minimum to 0.8 V maximum.
2. Both halves connected as shown.

FIGURE 10. Burn-in and life test circuits - Continued.

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Device type 04

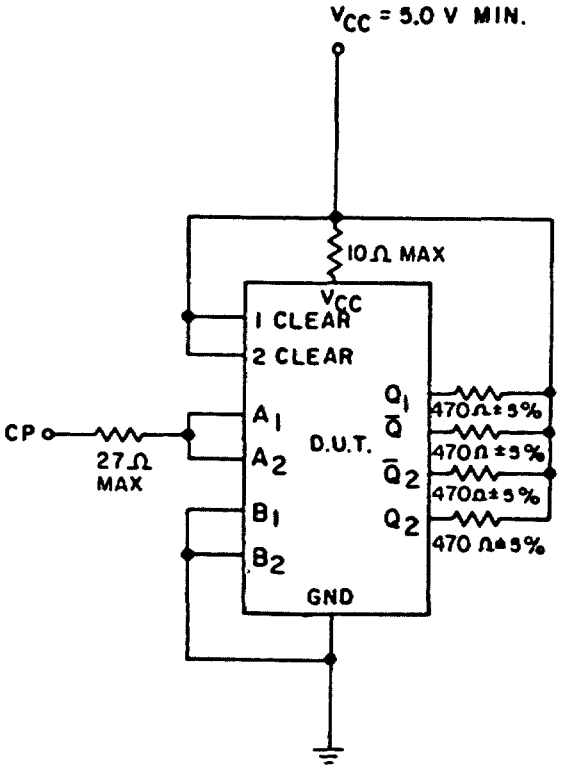


NOTE: CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IH} = 2.0 V minimum to 5.5 V maximum; V_{IL} = -0.5 V minimum to 0.8 V maximum.

FIGURE 10. Burn-in and life test circuits - Continued.

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Device type 05



NOTE: CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IH} = 2.0 V minimum to 5.5 V maximum; V_{IL} = -0.5 V minimum to 0.8 V maximum.

FIGURE 10. Burn-in and life test circuits - Continued.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥ 2.4 V, or low ≤ 0.4 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits		Unit
			Test no.	$\overline{Q}$	NC	A_1	A_2	B	Q	GND	NC	R_I	C_X	R_XC_X	NC	NC	V_{CC}		Min	Max	
1 T_C = +25°C	V_{OL}	3007 1/	1			0.8 V	0.8 V	0.8 V	16 mA	GND		4.5 V					4.5 V	Q		0.4	V
	V_{OL}	3007	2	16 mA		"	2.0 V	"		"		GND	GND				"	$\overline{Q}$		0.4	"
	V_{OH}	3006 1/	3	-.4 mA		"	0.8 V	"		"		4.5 V					"	$\overline{Q}$	2.4		"
	V_{OH}	3006	4			"	2.0 V	"	-.4 mA	"		GND	GND				"	Q	2.4		"
	V_{IC}		5			-12 mA				"							"	A_1		-1.5	"
	V_{IC}		6				-12 mA			"							"	A_2		-1.5	"
	V_{IC}		7					-12 mA		"							"	B		-1.5	"
	I_{IL1}	3009	8			0.4 V	5.5 V			"							5.5 V	A_1	-0.6	-1.6	mA
	I_{IL1}	3009	9			5.5 V	0.4 V			"							"	A_2	-0.6	-1.6	"
	I_{IL2}	3009	10			GND	GND	0.4 V		"		4.5 V					"	B	-1.4	-3.2	"
	I_{IH1}	3010	11			2.4 V	GND			"							"	A_1		40	µA
	I_{IH1}	"	12			GND	2.4 V			"							"	A_2		40	"
	I_{IH2}	"	13			5.5 V	GND			"							"	A_1		100	"
	I_{IH2}	"	14			GND	5.5 V			"							"	A_2		100	"
	I_{IH3}	"	15			5.5 V	5.5 V	2.4 V		"		4.5 V					"	B		80	"
	I_{IH4}	"	16			5.5 V	5.5 V	5.5 V		"		4.5 V					"	B		200	"
	I_{OS}	3011	17			GND	GND		GND	"		GND	GND	GND			"	Q	-20	-55	mA
	I_{OS}	3011	18	GND		"	"	GND		"		5.5 V					"	$\overline{Q}$	-20	-55	"
	I_{CC1}	3005	19			"	"	"		"		5.5 V					"	V_{CC}		25	"
	I_{CC2}	3005	20			"	"	"		"		GND	GND				"	V_{CC}		40	"
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
7 T_C = +25°C	Truth table test	3014	21	H		2.0 V	2.0 V	2/	L	GND		3/	4/	4/			5.0 V	6/			
		"	22	"		"	2.0 V	5/	"	"		"	"	"			"				
		"	23	"		"	0.8 V	5/	"	"		"	"	"			"				
		"	24	"		0.8 V	2.0 V	5/	"	"		"	"	"			"				
		"	25	"		0.8 V	2/	0.8 V	"	"		"	"	"			"				
		"	26	"		2/	0.8 V	"	"	"		"	"	"			"				
		"	27	"		2.0 V	2/	"	"	"		"	"	"			"				
		"	28	"		2/	2.0 V	"	"	"		"	"	"			"				
		"	29	"		2/	2/	"	"	"		"	"	"			"				
		"	30	"		2.0 V	2/	2.0 V	"	"		"	"	"			"				
		"	31	"		2/	2.0 V	"	"	"		"	"	"			"				
		"	32	"		2/	2/	"	"	"		"	"	"			"				
		"	33	7/		0.8 V	2.0 V	2/	8/	"		"	"	"			"				
		"	34	"		2.0 V	0.8 V	2/	"	"		"	"	"			"				
		"	35	"		2.0 V	5/	2.0 V	"	"		"	"	"			"				
		"	36	"		5/	2.0 V	"	"	"		"	"	"			"				
		"	37	"		5/	5/														
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																				

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.4 V, or low ≤ 0.4 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits		Unit
			Test no.	$\overline{Q}$	NC	A_1	A_2	B	Q	GND	NC	R_I	C_X	R_XC_X	NC	NC	V_{CC}		Min	Max	
9 T_C = +25°C	t_{PLH1}	3003	38			GND		IN	OUT	GND		3/	9/	9/			5.0 V	B to Q	15	59	ns
	t_{PHL1}	(Fig. 4)	39	OUT		GND		IN										B to $\overline{Q}$	20	69	
	t_{PLH2}		40			5.0 V	IN	5.0 V	OUT									A_2 to Q	25	74	
	t_{PHL2}		41	OUT		5.0 V	IN	5.0 V										A_2 to $\overline{Q}$	30	84	
	$t_{P(OUT)1}$	3003	42			GND	GND	IN	OUT				10/	10/				Q	70	150	
	$t_{P(OUT)2}$	(Fig. 5)	43										11/	11/ 13/				Q	20	50	
	$t_{P(OUT)3}$		44										12/	12/ 13/				Q	600	825	
	$t_{P(OUT)4}$		45															Q	5.5	8	µs
10 T_C = +125°C	t_{PLH1}	3003	46									3/	9/	9/				B to Q	15	75	ns
	t_{PHL1}	(Fig. 4)	47	OUT														B to $\overline{Q}$	20	87	
	t_{PLH2}		48			5.0 V	IN	5.0 V	OUT									A_2 to Q	25	93	
	t_{PHL2}		49	OUT		5.0 V	IN	5.0 V										A_2 to $\overline{Q}$	30	106	
	$t_{P(OUT)1}$	3003	50			GND	GND	IN	OUT				10/	10/				Q	70	150	
	$t_{P(OUT)2}$	(Fig. 5)	51										11/	11/ 13/				Q	20	50	
	$t_{P(OUT)3}$		52										12/	12/ 13/				Q	600	825	
	$t_{P(OUT)4}$		53															Q	5.5	8	µs
11	Same tests, terminal conditions and limits as subgroup 10, except T_C = -55°C.																				

1/ For circuit D, test numbers 2 and 4, terminal A_2 shall be 0.8 V and B shall be 2.0 V.
2/ In transition from low level to high level.
3/ R_I connected to V_{CC}.
4/ R_XC_X and C_X are open.
5/ In transition from high level to low level.
6/ Output voltages for subgroups 7 and 8: H ≥ 1.5 V, L ≤ 1.5 V.
7/ One low logic level pulse.
8/ One high logic level pulse.
9/ C_X connected to R_XC_X through an 80 pF capacitor.
10/ C_X connected to R_XC_X through a 15 pF capacitor which includes stray, probe, and jig capacitance.
11/ C_X connected to R_XC_X through a 100 pF capacitor.
12/ C_X connected to R_XC_X through a 1,000 pF capacitor.
13/ R_XC_X connected to V_{CC} through a 10 kΩ resistor.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥ 2.4 V, or low ≤ 0.4 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits		Unit
			Test no.	A_1	A_2	B_1	B_2	Clear	$\overline{Q}$	GND	Q	R_I	NC	C_X	NC	R_XC_X	V_{CC}		Min	Max	
1 T_C = +25°C	V_{OL}	3007	1	0.8 V		2.0 V	0.8 V	2.0 V	16 mA	GND	16 mA	4.5 V		GND			4.5 V	Q		0.4	V
	V_{OL}	3007	2									GND						$\overline{Q}$		0.4	
	V_{OH}	3006	3	0.8 V		2.0 V			-.8 mA		-.8 mA	GND		GND				Q	2.4		
	V_{OH}	3006	4									4.5 V						$\overline{Q}$	2.4		
	V_{IC}		5	-12 mA														A_1		-1.5	
			6		-12 mA													A_2			
			7			-12 mA												B_1			
			8				-12 mA											B_2			
			9					-12 mA										Clear			
	I_{IL1}	3009	10	0.4 V	5.5 V												5.5 V	A_1	-0.7	-1.6	mA
			11	5.5 V	0.4 V													A_2			
			12		GND	0.4 V	5.5 V	5.5 V										B_1			
			13			5.5 V	0.4 V	5.5 V										B_2			
	I_{IL2}		14			5.5 V	5.5 V	0.4 V										Clear	-1.4	-3.2	
	I_{IH1}	3010	15	2.4 V														A_1		40	μA
			16	GND	2.4 V													A_2			
			17	5.5 V	5.5 V	2.4 V	GND	GND										B_1			
			18		5.5 V	GND	2.4 V	GND										B_2			
	I_{IH2}		19		GND													A_1		100	
			20	GND	5.5 V													A_2			
			21	5.5 V		5.5 V	GND	GND										B_1			
			22			GND	5.5 V	GND										B_2			
	I_{IH3}		23			GND	GND	2.4 V										Clear		50	
	I_{IH4}		24			GND		5.5 V										Clear		200	
	I_{OS}	3011	25	GND		5.5 V					GND	GND		GND				Q	-10	-40	mA
	I_{OS}	3011	26						GND			5.5 V						$\overline{Q}$	-10	-40	
	I_{CC1}	3005	27	5.5 V	5.5 V	GND						5.5 V						V_{CC}		28	
	I_{CC2}	3005	28	GND	GND	5.5 V	5.5 V					GND		0.8 V				V_{CC}		28	
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
7 T_C = +25°C	Truth table test	3014	29					GND	H	GND	L					1/	5.0 V	6/			
			30	5.0 V	5.0 V	2/	2/	5.0 V													
			31			5.0 V	2/														
			32			2/	5.0 V														
			33	GND	3/	5.0 V															
			34	3/	GND	5.0 V															
			35	GND		2/			4/		5/										
			36	GND		5.0 V	2/														
			37		GND	2/	5.0 V														
			38		GND	5.0 V	2/														
			39	5.0 V	3/		5.0 V														
			40	3/	5.0 V																
			41	3/	3/																
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																				

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.4 V, or low ≤ 0.4 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Test no.	1 A_1	2 A_2	3 B_1	4 B_2	5 Clear	6 $\overline{Q}$	7 GND	8 Q	9 R_1	10 NC	11 C_X	12 NC	13 R_XC_X	14 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9	t_{PLH1}	3003	42	IN	5.0 V	5.0 V	5.0 V	5.0 V		GND	OUT			//		// 8/	5.0 V	A_1 to Q	7	37	ns
T_C = +25°C	t_{PLH2}	(Fig. 6)	43	GND	GND	IN	"	5.0 V		"	OUT			"		"	"	B_1 to Q	"	32	"
	t_{PLH3}	"	44	9/	5.0 V	5.0 V	"	IN	OUT	"				"		"	"	Clear to $\overline{Q}$	"	43	"
	t_{PHL1}	"	45	IN	5.0 V	5.0 V	"	5.0 V	OUT	"				"		"	"	A_1 to $\overline{Q}$	"	43	"
	t_{PHL2}	"	46	GND	GND	IN	"	5.0 V	OUT	"				"		"	"	B_1 to $\overline{Q}$	"	43	"
	t_{PHL3}	"	47	9/	5.0 V	5.0 V	"	IN		"	OUT			"		"	"	Clear to Q	"	30	"
	$t_{W(H)}$ 11/	"	48	IN	"	"	"	5.0 V		"	"					1/	"	Q		65	"
	$t_{W(H)}$ 11/	"	49	IN	"	"	"	"		"	"			10/		10/ 1/	"	Q		95	"
	t_W	"	50	GND	GND	IN	"	"		"	"			//		// 8/	"	Q	3.09	3.76	µs
10	t_{PLH1}	"	51	IN	5.0 V	5.0 V	"	"		"	"			"		"	"	A_1 to Q	7	48	ns
T_C = +125°C	t_{PLH2}	"	52	GND	GND	IN	"	"		"	"			"		"	"	B_1 to Q	"	41	"
	t_{PLH3}	"	53	9/	5.0 V	5.0 V	"	IN	OUT	"	"			"		"	"	Clear to $\overline{Q}$	"	56	"
	t_{PHL1}	"	54	IN	5.0 V	5.0 V	"	5.0 V	OUT	"				"		"	"	A_1 to $\overline{Q}$	"	56	"
	t_{PHL2}	"	55	GND	GND	IN	"	5.0 V	OUT	"				"		"	"	B_1 to $\overline{Q}$	"	51	"
	t_{PHL3}	"	56	9/	5.0 V	5.0 V	"	IN		"	OUT			"		"	"	Clear to Q	"	39	"
	$t_{W(H)}$ 11/	"	57	IN	"	"	"	5.0 V		"	"					1/	"	Q		75	"
	$t_{W(H)}$ 12/	"	58	IN	"	"	"	"		"	"			10/		10/	"	Q		105	"
	t_W	"	59	GND	GND	IN	"	"		"	"			//		// 8/	"	Q	2.60	4.15	µs
11	Same tests, terminal conditions and limits as for subgroup 10 except T_C = -55°C.																				

1/ R_XC_X connected to V_{CC} through a 5 kΩ resistor.
2/ In transition from low level to high level.
3/ In transition from high level to low level.
4/ One low logic level pulse.
5/ One high logic level pulse.
6/ Output voltages for subgroups 7 and 8: H > 1.5 V, L < 1.5 V.
7/ C_X connected to R_XC_X through a 1,000 pF capacitor.
8/ R_XC_X connected to V_{CC} through a 10 kΩ resistor.
9/ Device must be triggered before t_{PHL3} can be measured (see figure 6).
10/ C_X connected to R_XC_X through a 15 pF capacitor which includes stray, probe, and jig capacitance.
11/ This test shall be performed for bench setup only.
12/ This test shall be performed for automatic test equipment only.

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pin not designated may be high ≥ 2.4 V, or low ≤ 0.4 V, or open).

Subgroup	Symbol	MIL-STD-883-method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits		Unit
			Test no.	A_1	B_1	Clear 1	$\overline{Q}_1$	Q_2	C_{X2}	R_XC_{X2}	GND	A_2	B_2	Clear 2	$\overline{Q}_2$	Q_1	C_{X1}	R_XC_{X1}	V_{CC}		Min	Max	
1 T_C = +25°C	V_{OL}	3007	1	2.0 V	0.8 V	GND	GND				GND					16 mA			4.5 V	$\overline{Q}_1$		0.4	V
			2	0.8 V	2.0 V	2.0 V				1/		2.0 V	0.8 V	GND			2/	1/		Q_1			
			3				16 mA	16 mA												$\overline{Q}_2$			
			4						2/			0.8 V	2.0 V	2.0 V	16 mA					Q_2			
	V_{OH}	3006	5	2.0 V	0.8 V	GND	-.8 mA													$\overline{Q}_1$	2.4		
			6	0.8 V	2.0 V	2.0 V										-.8 mA	2/	1/		Q_1			
			7							1/		2.0 V	0.8 V	GND	-.8 mA					$\overline{Q}_2$			
			8					-.8 mA	2/			0.8 V	2.0 V	2.0 V						Q_2			
	V_{IC}		9	-12 mA																A_1		-1.5	
			10		-12 mA															B_1			
			11			-12 mA														Clear 1			
			12									-12 mA								A_2			
			13										-12 mA							B_2			
			14											-12 mA						Clear 2			
	I_{IL1}	3009	15	0.4 V															5.5 V	A_1	-0.7	-1.6	mA
			16	GND	0.4 V	5.5 V														B_1			
			17									0.4 V								A_2			
			18									GND	0.4 V	5.5 V						B_2			
	I_{IL2}		19	GND	5.5 V	0.4 V														Clear 1	3/	3/	
	I_{IL2}		20									GND	5.5 V	0.4 V						Clear 2			
	I_{IH1}	3010	21	2.4 V																A_1		40	μA
			22	5.5 V	2.4 V	GND														B_1			
			23									2.4 V								A_2			
			24									5.5 V	2.4 V	GND						B_2			
	I_{IH2}		25	5.5 V																A_1		100	
			26	5.5 V	5.5 V	GND														B_1			
			27									5.5 V								A_2			
			28									5.5 V	5.5 V	GND						B_2			
	I_{IH3}		29	5.5 V	GND	2.4 V														Clear 1		50	
	I_{IH3}		30									5.5 V	GND	2.4 V						Clear 2		80	
	I_{IH4}		31	5.5 V	GND	5.5 V														Clear 1		200	
	I_{IH4}		32									5.5 V	GND	5.5 V						Clear 2		200	
	I_{OS}	3011	33	GND	4/	5.5 V										GND	2/			$\overline{Q}_1$	-10	-40	mA
			34	GND	GND	GND	GND											1/		Q_1			
			35					GND	2/			GND	4/	5.5 V						$\overline{Q}_2$			
			36							1/		GND	GND	GND	GND					Q_2			
	I_{CC1}	3005	37	2.4 V	GND	2.4 V				1/		2.4 V	GND	2.4 V				1/		V_{CC}		66	
	I_{CC2}	3005	38	2.4 V	GND	2.4 V			2/			2.4 V	GND	2.4 V			2/			V_{CC}		66	
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = +25°C	Truth table test	3014	39			GND	H	L		1/	GND			GND	H	L		1/	5.0 V	7/			
			40	5.0 V	4/	5.0 V	H	L				5.0 V	4/	5.0 V	H	L							
			41	5/	GND		H	L				5/	GND		H	L							
			42	GND	4/		6/	8/				GND	4/		6/	8/							
			43	5/	5.0 V		6/	8/				5/	5.0 V		6/	8/							

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pin not designated may be high ≥ 2.4 V, or low ≤ 0.4 V, or open).

Subgroup	Symbol	MIL-STD-883-method	Cases E, F Test no.	1 A_1	2 B_1	3 Clear 1	4 $\overline{Q}_1$	5 Q_2	6 C_{X2}	7 R_XC_{X2}	8 GND	9 A_2	10 B_2	11 Clear 2	12 $\overline{Q}_2$	13 Q_1	14 C_{X1}	15 R_XC_{X1}	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						
9 T_C = +25°C	t_{PLH1}	3003 (Fig. 7)	44	IN	5.0 V	5.0 V		OUT	9/	9/ 10/	GND	IN	5.0 V	5.0 V		OUT	9/	9/ 10/	5.5 V	A_1 to $\overline{Q}_1$	7	37	ns
	t_{PLH1}		45																	A_2 to Q_2		37	
	t_{PLH2}		46	GND	IN	5.0 V		OUT	9/	9/ 10/		GND	IN	5.0 V		OUT	9/	9/ 10/		B_1 to $\overline{Q}_1$		32	
	t_{PLH2}		47																	B_2 to $\overline{Q}_2$		32	
	t_{PLH3}		48	11/	5.0 V	IN	OUT										9/	9/ 10/		Clear 1 to $\overline{Q}_1$		44	
	t_{PLH3}		49						9/	9/ 10/		11/	5.0 V	IN	OUT					Clear 2 to $\overline{Q}_2$		44	
	t_{PHL1}		50	IN	5.0 V	5.0 V	OUT										9/	9/ 10/		A_1 to $\overline{Q}_1$		44	
	t_{PHL1}		51						9/	9/ 10/		IN	5.0 V	5.0 V	OUT					A_2 to $\overline{Q}_2$		44	
	t_{PHL2}		52	GND	IN	5.0 V	OUT										9/	9/ 10/		B_1 to $\overline{Q}_1$		40	
	t_{PHL2}		53						9/	9/ 10/		GND	IN	5.0 V	OUT					B_2 to $\overline{Q}_2$		40	
	t_{PHL3}		54	11/	5.0 V	IN										OUT	9/	9/ 10/		Clear 1 to Q_1		32	
	t_{PHL3}		55					OUT	9/	9/ 10/		11/	5.0 V	IN						Clear 2 to Q_2		32	
	$t_{W(H\|H)}$ 13/		56	IN	5.0 V	5.0 V										OUT		1/		Q_1	35	65	
	$t_{W(H\|H)}$ 13/		57					OUT		1/		IN	5.0 V	5.0 V						Q_2		65	
	$t_{W(H\|H)}$ 14/		58	IN	5.0 V	5.0 V										OUT	12/	1/		Q_1		95	
	$t_{W(H\|H)}$ 14/		59					OUT	12/	1/		IN	5.0 V	5.0 V						Q_2		95	
	t_W		60	GND	IN	5.0 V		OUT	9/	9/ 10/		GND	IN	5.0 V		OUT	9/	9/ 10/		Q_1	3.08	3.76	ns
	t_W		61																	Q_2	3.08	3.76	
10 T_C = +125°C	t_{PLH1}		62	IN	5.0 V	5.0 V		OUT	9/	9/ 10/		IN	5.0 V	5.0 V		OUT	9/	9/ 10/		A_1 to $\overline{Q}_1$	7	48	ns
	t_{PLH1}		63																	A_2 to Q_2		48	
	t_{PLH2}		64	GND	IN	5.0 V		OUT	9/	9/ 10/		GND	IN	5.0 V		OUT	9/	9/ 10/		B_1 to $\overline{Q}_1$		41	
	t_{PLH2}		65																	B_2 to $\overline{Q}_2$		41	
	t_{PLH3}		66	11/	5.0 V	IN	OUT										9/	9/ 10/		Clear 1 to $\overline{Q}_1$		56	
	t_{PLH3}		67						9/	9/ 10/		11/	5.0 V	IN	OUT					Clear 2 to $\overline{Q}_2$		56	
	t_{PHL1}		68	IN	5.0 V	5.0 V	OUT										9/	9/ 10/		A_1 to $\overline{Q}_1$		56	
	t_{PHL1}		69						9/	9/ 10/		IN	5.0 V	5.0 V	OUT					A_2 to $\overline{Q}_2$		56	
	t_{PHL2}		70	GND	IN	5.0 V	OUT										9/	9/ 10/		B_1 to $\overline{Q}_1$		51	
	t_{PHL2}		71						9/	9/ 10/		GND	IN	5.0 V	OUT					B_2 to $\overline{Q}_2$		51	
	t_{PHL3}		72	11/	5.0 V	IN										OUT	9/	9/ 10/		Clear 1 to Q_1		39	
	t_{PHL3}		73					OUT	9/	9/ 10/		11/	5.0 V	IN						Clear 2 to Q_2		39	

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pin not designated may be high ≥ 2.4 V, or low ≤ 0.4 V, or open).

Subgroup	Symbol	MIL-STD-883-method	Cases E, F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminals	Test limits Min	Test limits Max	Unit
				A_1	B_1	Clear 1	$\overline{Q}_1$	Q_2	C_{X2}	R_XC_X2	GND	A_2	B_2	Clear 2	$\overline{Q}_2$	Q_1	C_{X1}	R_XC_X1	V_{CC}				
10 T_C = 125°C	$t_{W(H)}$ 13/	3003 (Fig. 7)	74	IN	5.0 V	5.0 V					GND					OUT		1/	5.0 V	Q_1	35	75	ns
	$t_{W(H)}$ 14/	"	75					OUT		1/	"	IN	5.0 V	5.0 V					"	Q_2	"	75	"
	$t_{W(L)}$ 13/	"	76	IN	5.0 V	5.0 V					"					OUT	11/	1/	"	Q_1	"	195	"
	$t_{W(L)}$ 14/	"	77					OUT	11/	1/	"	IN	5.0 V	5.0 V					"	Q_2	"	105	"
	t_H	"	78	GND	IN	5.0 V					"					OUT	9/	9/ 10/	"	Q_1	2.60	4.15	µs
	t_d	"	79					OUT	9/	9/ 10/	"	GND	IN	5.0 V					"	Q_2	2.60	4.15	µs
11	Same tests, terminal conditions and limits as subgroup 10, except T_C = -55°C.																						

1/ R_XC_X connected to V_{CC} through a 5 kΩ resistor.
2/ C_X connected to GND or 0.8 V.
3/ For schematic circuits A, B, and C, the minimum and maximum limits shall be -1.4 mA and -3.2 mA, respectively. For schematic circuit D, the minimum and maximum limits shall be -0.7 mA and -1.4 mA, respectively.
4/ In transition from low level to high level.
5/ In transition from high level to low level.
6/ One low logic level pulse.
7/ Output voltages for subgroups 7 and 8: H > 1.5 V, L < 1.5 V.
8/ One high logic level pulse.
9/ C_X connected to R_XC_X through a 1,000 pF capacitor.
10/ R_XC_X connected to V_{CC} through a 10 kΩ resistor.
11/ Device must be triggered before t_{PLH3} and t_{PHL3} can be measured (see figure 7).
12/ C_X connected to R_XC_X through a 15 pF capacitor which includes stray, probe, and jig capacitance.
13/ This test shall be performed for bench setup only.
14/ This test shall be performed for automatic test equipment only.

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TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated may be high > 2.4 V, or low < 0.4 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Min	Test limits Max	Unit
				A_1	A_2	B_1	B_2	NC	$\overline{Q}$	GND	Q	NC	NC	C_X	NC	R_XC_X	V_{CC}				
1 T_C = $T_{25°C}$	V_{OL}	3007	1				0.8 V			GND	10 mA					1/	4.5 V	$\overline{Q}$		0.4	V
	V_{OL}	3007	2		0.8 V	2.0 V	2.0 V		10 mA	"				GND		$\overline{1}$/	"	Q		0.4	"
	V_{OH}	3006	3		0.8 V	2.0 V	2.0 V			"	-.72 mA					1/	"	$\overline{Q}$	2.4		"
	V_{OH}	3006	4				0.8 V		-.72 mA	"				GND		$\overline{1}$/	"	Q	2.4		"
	V_{IC}		5	-12 mA						"							"	A_1		-1.5	"
			6		-12 mA					"							"	A_2		"	"
			7			-12 mA				"							"	B_1		"	"
			8				-12 mA			"							"	B_2		"	"
	I_{IL1}	3009	9	0.4 V	5.5 V					"							5.5 V	A_1	-0.7	-1.6	mA
		"	10	5.5 V	0.4 V					"							"	A_2	"	"	"
		"	11		GND	0.4 V	5.5 V			"							"	B_1	"	"	"
		"	12		GND	5.5 V	0.4 V			"							"	B_2	"	"	"
	I_{IH1}	3010	13	2.4 V	GND					"							"	A_1		40	uA
		"	14	GND	2.4 V					"							"	A_2		"	"
		"	15	5.5 V	5.5 V	2.4 V	GND			"							"	B_1		"	"
		"	16	5.5 V	5.5 V	GND	2.4 V			"							"	B_2		"	"
	I_{IH2}	"	17	5.5 V	GND					"							"	A_1		100	"
		"	18	GND	5.5 V					"							"	A_2		"	"
		"	19	5.5 V	5.5 V	5.5 V	GND			"							"	B_1		"	"
		"	20	5.5 V	5.5 V	GND	5.5 V			"							"	B_2		"	"
	I_{OS}	3011	21		GND	5.5 V	5.5 V		GND	"						1/	"	$\overline{Q}$	-10	-40	mA
	I_{OS}	3011	22				GND			"	GND			GND		$\overline{1}$/	"	Q	-10	-40	"
	I_{CC}	3005	23	GND	GND	5.5 V	5.5 V			"						1/		V_{CC}		25	"
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
7 T_C = $T_{25°C}$	Truth table test	3014	24	5.0 V	5.0 V				H	GND	L					2/	5.0 V	6/			
		"	25			GND			"	"	"					"	"				
		"	26				GND		"	"	"					"	"				
		"	27	GND		5.0 V	5.0 V		"	"	"					"	"				
		"	28		GND	5.0 V	5.0 V		"	"	"					"	"				
		"	29	GND		3/	5.0 V		4/	"	5/					"	"				
		"	30	GND		5.0 V	3/		"	"	"					"	"				
		"	31		GND	3/	5.0 V		"	"	"					"	"				
		"	32		GND	5.0 V	3/		"	"	"					"	"				
		"	33	5.0 V	7/		5.0 V		"	"	"					"	"				
		"	34	7/	5.0 V		5.0 V		"	"	"					"	"				
		"	35	$\overline{7}$/	7/		5.0 V		"	"	"					"	"				
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																				
9 T_C = $T_{25°C}$	t_{PLH1}	3003	36	IN	5.0 V	5.0 V	5.0 V			GND	OUT					2/	5.0 V	A_1 to Q	7	37	ns
	t_{PLH2}	(Fig. 8)	37	GND	GND	IN	"			"	OUT					"	"	B_1 to Q	"	32	"
	t_{PHL1}	"	38	IN	5.0 V	5.0 V	"		OUT	"						"	"	A_1 to $\overline{Q}$	"	44	"
	t_{PHL2}	"	39	GND	GND	IN	"		OUT	"						"	"	B_1 to $\overline{Q}$	"	40	"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pin not designated may be high ≥ 2.4 V, or low ≤ 0.4 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Min	Test limits Max	Unit
				A_1	A_2	B_1	B_2	NC	$\overline{Q}$	GND	Q	NC	NC	C_X	NC	R_XC_X	V_{CC}				
9 T_C = +25°C	$t_{W(HIN)}$ 8/	3003 (Fig. 8)	40	IN	5.0 V	5.0 V	5.0 V			GND	OUT					2/	5.0 V	Q	25	75	ns
	$t_{W(HIN)}$ 9/	"	41	IN	"	"	"			"	"			10/		2/	"	Q	25	105	ns
	t_W	"	42	7/	"	"	"			"	"			11/		2/ 1/	"	Q	3.08	3.76	µs
10 T_C = +125°C	t_{PLH1}	"	43	IN	"	"	"			"	"					2/	"	A_1 to Q	7	48	ns
	t_{PLH2}	"	44	GND	GND	IN	"			"	"					"	"	B_1 to Q	"	41	"
	t_{PHL1}	"	45	IN	"	"	"		OUT	"						"	"	A_1 to $\overline{Q}$	"	56	"
	t_{PHL2}	"	46	GND	GND	IN	"		OUT	"						"	"	B_1 to $\overline{Q}$	"	51	"
	$t_{W(HIN)}$ 8/	"	47	IN	5.0 V	5.0 V	"			"	OUT					"	"	Q	25	95	"
	$t_{W(HIN)}$ 9/	"	48	IN	"	"	"			"	"			10/		"	"	Q	25	125	"
	t_W	"	49	7/	"	"	"			"	"			11/		11/ 1/	"	Q	2.60	4.10	µs
11	Same tests, terminal conditions and limits as subgroup 10, except T_C = -55°C.																				

1/ R_XC_X connected to V_{CC} through a 10 kΩ resistor.
2/ R_XC_X connected to V_{CC} through a 5 kΩ resistor.
3/ In transition from low level to high level.
4/ One low logic level pulse.
5/ One high logic level pulse.
6/ Output voltages for subgroups 7 and 8: H > 1.5 V, L < 1.5.
7/ In transition from high level to low level.
8/ This test shall be performed for bench setup only.
9/ This test shall be performed for automatic test equipment only.
10/ C_X connected to R_XC_X through a 15 pF capacitor which includes stray, probe, and jig capacitance.
11/ C_X connected to R_XC_X through a 1,000 pF capacitor.

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TABLE III. Group A inspection for device type 05.
Terminal conditions (pins not designated may be high ≥ 2.4 V, or low ≤ 0.4 V, or open)

Subgroup	Symbol	MIL-STD-883-method	Cases E, F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				C_{X1}	R_XC_{X1}	Clear 1	B_1	A_1	Q_1	$\overline{Q}_1$	GND	$\overline{Q}_2$	Q_2	A_2	B_2	Clear 2	R_XC_{X2}	C_{X2}	V_{CC}				
1 T_C = +25°C	V_{OL}	3007	1 2 3 4		0.9 V	0.8 V 2.0 V	GND	1/	10 mA	10 mA	GND	10 mA	10 mA	1/	GND	0.8 V 2.0 V	0.9 V		4.5 V	Q_1 $\overline{Q}_1$ Q_2 $\overline{Q}_2$		0.4	V
	V_{OH}	3006	5 6 7 8		0.9 V	2.0 V 0.8 V	GND	1/	-.96 mA	-.96 mA		-.96 mA	-.96 mA	1/	GND	2.0 V 0.8 V	0.9 V			Q_1 $\overline{Q}_1$ Q_2 $\overline{Q}_2$	2.4		V
	V_{IC}		9 10 11 12 13 14			-12 mA	-12 mA	-12 mA						-12 mA	-12 mA	-12 mA					Clear 1 B_1 A_1 Clear 2 B_2 A_2		-1.5	V
	I_{IL}	3009	15 16 17 18 19 20			0.4 V	0.4 V	0.4 V						0.4 V	0.4 V	0.4 V			5.5 V	Clear 1 B_1 A_1 Clear 2 B_2 A_2	-0.7	-1.6	mA
	I_{IH1}	3010	21 22 23 24 25 26			2.4 V	2.4 V	2.4 V						2.4 V	2.4 V	2.4 V				Clear 1 B_1 A_1 Clear 2 B_2 A_2		40	µA
	I_{IH2}		27 28 29 30 31 32			5.5 V	5.5 V	5.5 V						5.5 V	5.5 V	5.5 V				Clear 1 B_1 A_1 Clear 2 B_2 A_2		100	
	I_{OS}	3011	33 34 35 36		0.9 V	2.0 V 0.8 V	GND	1/	0.5 V	0.5 V		0.5 V	0.5 V	1/	GND	2.0 V 0.8 V	0.9 V		6.0 V	Q_1 $\overline{Q}_1$ Q_2 $\overline{Q}_2$	-10	-40	mA
	I_{CC} I_{CC}	3005 3005	37 38	GND	GND		GND	GND						GND	GND		GND	GND	5.5 V 5.5 V	V_{CC} V_{CC}		52 52	
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = +25°C	Truth table test	3014	39 40 41 42 43		2/	GND 5.0 V	3/ 5.0 V 3/ GND	GND 1/ 5.0 V 1/	L L 4/ 4/	H H H 5/ 5/	GND	H H H 5/ 5/	L L 4/ 4/	GND 1/ 5.0 V 1/	3/ 5.0 V 3/ GND	GND 5.0 V	2/		5.0 V	6/			

See footnotes at end of device type 05.

MIL-M-38510/12G

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.4 V, or low ≤ 0.4 V, or open).

Subgroup	Symbol	MIL-STD-883-method	Cases E, F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				C_{X1}	R_XC_{X1}	Clear 1	B_1	A_1	Q_1	$\overline{Q}_1$	GND	$\overline{Q}_2$	Q_2	A_2	B_2	Clear 2	R_XC_{X2}	C_{X2}	V_{CC}				
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						
9 T_C = +25°C	t_{PLH1}	3003 (Fig. 9)	44		2/	5.0 V	GND	IN	OUT		GND		OUT	IN	GND	5.0 V	2/		5.0 V	A_1 to $\overline{Q}_1$	7	40	ns
	t_{PLH1}	"	45																	A_2 to $\overline{Q}_2$		40	
	t_{PLH2}	"	46		2/	5.0 V	IN	5.0 V	OUT		"		OUT	5.0 V	IN	5.0 V	2/			B_1 to $\overline{Q}_1$	"	37	"
	t_{PLH2}	"	47																	B_2 to $\overline{Q}_2$	"	37	"
	t_{PLH3}	"	48		2/	IN	GND	7/		OUT	"								"	Clear 1 to $\overline{Q}_1$	"	44	"
	t_{PLH3}	"	49									OUT		7/	GND	IN	2/			Clear 2 to $\overline{Q}_2$	"	44	"
	t_{PHL1}	"	50		2/	5.0 V	GND	IN		OUT	"								"	A_1 to $\overline{Q}_1$	"	48	"
	t_{PHL1}	"	51								"	OUT		IN	GND	5.0 V	2/		"	A_2 to $\overline{Q}_2$	"	48	"
	t_{PHL2}	"	52		2/	5.0 V	IN	5.0 V		OUT	"								"	B_1 to $\overline{Q}_1$	"	46	"
	t_{PHL2}	"	53								"	OUT		5.0 V	IN	5.0 V	2/		"	B_2 to $\overline{Q}_2$	"	46	"
	t_{PHL3}	"	54		2/	IN	GND	7/	OUT		"								"	Clear 1 to $\overline{Q}_1$	"	32	"
	t_{PHL3}	"	55										OUT	7/	GND	IN	2/			Clear 2 to $\overline{Q}_2$	"	32	"
	$t_{W(H)d}$ 8/	"	56		2/	5.0 V	GND	IN	OUT		"								"	Q_1	35	93	"
	$t_{W(H)U}$ 8/	"	57										OUT	IN	GND	5.0 V	2/			Q_2	"	93	"
	$t_{W(H)H}$ 9/	"	58	10/	2/	5.0 V	GND	IN	OUT		"								"	Q_1	"	120	"
	$t_{d(H)H}$ 9/	"	59										OUT	IN	GND	5.0 V	2/	10/	"	Q_2	"	128	"
	t_W	"	60	11/	11/ 12/	5.0 V	GND	1/	OUT		"								"	Q_1	2.60	3.76	µs
	t_d	"	61										OUT	1/	GND	5.0 V	11/ 12/	11/	"	Q_2	2.60	3.76	µs
10 T_C = +125°C	t_{PLH1}	"	62		2/	5.0 V	GND	IN	OUT		"								"	A_1 to $\overline{Q}_1$	7	54	ns
	t_{PLH1}	"	63										OUT	IN	GND	5.0 V	2/			A_2 to $\overline{Q}_2$	"	54	"
	t_{PLH2}	"	64		2/	5.0 V	IN	5.0 V	OUT		"								"	B_1 to $\overline{Q}_1$	"	51	"
	t_{PLH2}	"	65										OUT	5.0 V	IN	5.0 V	2/			B_2 to $\overline{Q}_2$	"	51	"
	t_{PLH3}	"	66		2/	IN	GND	7/		OUT	"								"	Clear 1 to $\overline{Q}_1$	"	56	"
	t_{PLH3}	"	67									OUT		7/	GND	IN	2/			Clear 2 to $\overline{Q}_2$	"	56	"
	t_{PHL1}	"	68		2/	5.0 V	GND	IN		OUT	"								"	A_1 to $\overline{Q}_1$	"	61	"
	t_{PHL1}	"	69								"	OUT		IN	GND	5.0 V	2/		"	A_2 to $\overline{Q}_2$	"	61	"
	t_{PHL2}	"	70		2/	5.0 V	IN	5.0 V		OUT	"								"	B_1 to $\overline{Q}_1$	"	58	"
	t_{PHL2}	"	71								"	OUT		5.0 V	IN	5.0 V	2/		"	B_2 to $\overline{Q}_2$	"	58	"

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.4 V, or low ≤ 0.4 V, or open).

Subgroup	Symbol	MIL-STD-883-method	Cases E, F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				C_X1	R_XC_X1	Clear 1	B_1	A_1	Q_1	$\overline{Q}_1$	GND	$\overline{Q}_2$	Q_2	A_2	B_2	Clear 2	R_XC_X2	C_X2	V_{CC}				
10 T_A = +125°C	t_{PHL3}	3003 (Fig. 9)	72		2/	IN	GND	1/	OUT		GND								5.0 V	Clear 1 to Q_1	7	39	ns
	t_{PHL3}	"	73				"				"		OUT	1/	GND	IN	2/		"	Clear 2 to Q_2	7	39	"
	$t_{W(MIN)}$ 8/	"	74		2/	5.0 V	GND	IN	OUT		"								"	Q_1	35	108	"
	$t_{W(MIN)}$ 8/	"	75								"		OUT	IN	GND	5.0 V	9/		"	Q_2	"	108	"
	$t_{W(MIN)}$ 9/	"	76	10/	2/	5.0 V	GND	IN	OUT		"								"	Q_1	"	138	"
	$t_{W(MIN)}$ 9/	"	77								"		OUT	IN	GND	5.0 V	2/	10/	"	Q_2	"	138	"
	t_W	"	78	11/	11/ 12/	5.0 V	GND	1/	OUT		"								"	Q_1	2.60	3.91	µs
	t_W	"	79								"		OUT	1/	GND	5.0 V	11/ 12/	11/	"	Q_2	2.60	3.91	µs
11	Same tests, terminal conditions and limits as subgroup 10, except T_A = -55°C, and t_W max limit = 4.10 µs.																						

1/ In transition from high level to low level.
2/ R_XC_X connected to V_{CC} through a 5 kΩ resistor.
3/ In transition from low level to high level.
4/ One high level logic pulse.
5/ One low level logic pulse.
6/ Output voltages for subgroups 7 and 8: H > 1.5 V, L < 1.5 V.
7/ Device must be triggered before t_{PLH3} and t_{PHL3} can be measured (see figure 9).
8/ This test shall be performed for bench setup only.
9/ This test shall be performed for automatic test equipment only.
10/ C_X connected to R_XC_X through a 15 pF capacitor which includes stray, probe, and jig capacitance.
11/ C_X connected to R_XC_X through a 1,000 pF capacitor.
12/ R_XC_X connected to V_{CC} through a 10 kΩ resistor.

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6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential.
V_{IN}	Voltage level at an input terminal.
V_{IC}	Input clamp voltage.
I_{IN}	Current flowing into an input terminal.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead material and finish C (see 3.3). Longer lead lengths and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54121
02	54122
03	54123
04	9601
05	9602

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Air Force - 11, 19, 85, 99
Navy - OS, SH, TD
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-0833)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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